HISTORY OF PSYCHOLOGY

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History of Psychology

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1: Foundations of Psychology

Psychology as we know it didn't suddenly appear on the intellectual scene. It is impossible to say just when it began, or who was responsible for it. Instead, we can only point to a number of currents that take us from philosophy and the natural sciences into something recognizably psychological. This chapter looks at two of these "primordial" currents -- associationism as the beginnings of a cognitive theory, and the introduction of quantification in the forms of psychophysics and intelligence testing.

Associationism

Associationism is the theory that the mind is composed of elements -- usually referred to as sensations and ideas -- which are organized by means of various associations. Although the original idea can be found in Plato, it is Aristotle who gets the credit for elaborating on it. Aristotle counted four laws of association when he examined the processes of remembrance and recall:

1. **The law of contiguity**. Things or events that occur close to each other in space or time tend to get linked together in the mind. If you think of a cup, you may think of a saucer; if you think of making coffee, you may then think of drinking that coffee.

2. **The law of frequency**. The more often two things or events are linked, the more powerful will be that association. If you have an eclair with your coffee every day, and have done so for the last twenty years, the association will be strong indeed -- and you will be fat.

3. **The law of similarity**. If two things are similar, the thought of one will tend to trigger the thought of the other. If you think of one twin, it is hard not to think of the other. If you recollect one birthday, you may find yourself thinking about others as well.

4. **The law of contrast**. On the other hand, seeing or recalling something may also trigger the recollection of something completely opposite. If you think of the tallest person you know, you may suddenly recall the shortest one as well. If you are thinking about birthdays, the one that was totally different from all the rest is quite likely to come up.

Association, according to Aristotle, took place in the "**common sense**." It was in the common sense that the look, the feel, the smell, the taste of an apple, for example, came together to become the idea of an apple.

For 2000 years, these four laws were assumed to hold true. St. Thomas pretty much accepted it lock, stock, and barrel. No one, however, cared that much about association. It was seen as just a simple description of a commonplace occurrence. It was seen as the activity of passive reason, whereas the abstraction of principles or essences -- far more significant to philosophers -- was the domain of active reason.

During the enlightenment, philosophers began to become interested in the idea again, as a part of their studies of vision as well as their interest in epistemology. **Hobbes** understood complex experiences as being associations of simple experiences, which in turn were associations of sensations. The basic means of association, according to Hobbes, was coherence (continguity), and the basic strength factor was repetition (frequency).

John Locke, rejecting the possibility of innate ideas, made his entire system dependent on association of sensations into simple ideas. He did, however, distinguish between ideas of sensations and ideas of reflection, meaning active reason. Only by adding simple ideas of reflection to simple ideas of sensation could we derive complex ideas. He also suggested that complex emotions derived from pain and pleasure (simple ideas) associated with other ideas.

It was **David Hume** who really got into the issue. Recall that he saw all experiences as having no substantial reality behind them. So whatever coherence the world (or the self) seems to have is a matter of the simple application of these natural laws of association. He lists three:

1. **The law of resemblance** -- i.e. similarity.

2. **The law of contiguity**.

3. **The law of cause and effect** -- basically contiguity in time.

David Hartley (1705-1757) was an English physician who was responsible for making the idea of associationism popular, especially in a book called Observations of Man. His emphasis was on the law of contiguity (in time and space) and the law of frequency. But he added an idea he got from the famous Isaac Newton: This association was a matter of tuned "**vibrations**" within the nerves! His basic ideas are very similar to those of D. O. Hebb in the twentieth century.

James Mill (1773-1836) also elaborated on Hume's associationism. The elder Mill saw the mind as passively functioning by the law of contiguity, with the law of frequency and a law of **vividness** "stamping in" the association. His emphasis on the law of frequency as the key to learning makes his approach very similar to the behaviorists in the twentieth century. But he is most famous for being the father of...

John Stuart Mill

"That so few now dare to be eccentric marks the chief danger of the time."

John Stuart Mill was born May 20, 1806 in London. His father was James Mill, an historian, philosopher, and social theorist. His mother was Harriet Barrow, and seems to have had next to no influence on him! His father decided to use the principles of utilitarianism and associationism (in consultation with his good friend, Jeremy Bentham) to educate John "scientifically."

This seemed to work quite well: John began learning Greek at three, Latin at eight. At 14, he studied French, mathematics, and chemistry in France. At 16, he began working as a clerk for his father at India House, headquarters of the East India Company. By 18, John was publishing articles on utilitarian philosophy!

But at 20, he had a nervous breakdown, which he describes in his **Autobiography** (1873). He attributed it, no doubt rightly, to his rigid education.

In 1830, he met Harriet Taylor, a married woman. He remained loyal to her until her husband died 21 years later (!), at which point they married. Sadly, she died only seven years later.

During this time, he served as an examiner for the East India Company. He also served as a liberal member of Parliament from 1865 to 1868. ("Conservatives are not necessarily stupid, but most stupid people are conservatives.") He died at his home in Avignon, France, on May 8, 1873.

His best known work is **On Liberty**, published in 1859. His most important work as far as science and psychology are concerned is **A System of Logic**, first printed in 1843 and going through many more editions through the rest of the 1800's.

He began with the basics established by Hume, his father James Mill, and others:

1. A sensory impression leaves a mental representation (idea or image);
2. If two stimuli are presented together repeated, they create an association in the mind;
3. The intensity of such a pairing can serve the same function as repetition.

But he adds that associations can be more than the simple sum of their parts. They can have attributes or qualities different from the parts in the same way that water has different qualities than the hydrogen and oxygen that compose it. So J. S. Mill's associationism is more like "**mental chemistry**" than mental addition.

J. S. Mill agrees with Hume that all we can know about our world and ourselves is what we experience, but notes that generalization allows us to talk with some confidence about things beyond experience. And he believed that there are real causes for consistent phenomena!

This is often called **phenomenalism**. He defines matter, for example, as "the permanent possibility of sensation." This persepctive would have profound effects on 20th century logical positivism (Wittgenstein, Ayer, Schlick, Carnap, and others) who provided the philosophical foundation for most behaviorists.

He promotes a scientific method that focuses on **induction**: Generalizations from experiences lead to theory, from which we then develop alternative hypotheses; We go on to test these hypotheses by observation and experiment, the results of which allow us to improve theory, and so on. This circular notion of scientific progress is known as the **hypothetico-deductive method**. In this way we slowly build up laws of nature in which we can be increasingly confident. This method proved to be very popular among the scientists of his day.

He more specifically outlines five procedures for establishing causation. The simpler ones go like this:

1. **The method of agreement**: If a phenomenon occurs in two different situations, and those two situations have only one thing in common, that "thing" is the cause (or effect) of the phenomenon.

2. **The method of differences**: If a phenomenon occurs in one situation but not in another, and those two situations have everything in common except for one thing, then that "thing" is the cause (or effect) of the phenomenon.

3. **The method of concomitant variations**: If one phenomenon varies consistently with the variations of another phenomenon, one is the cause or effect, or is otherwise involved in the causation, of the other. This, of course, is the foundation for correlation which,

although it cannot establish the direction of causality, does indicate some causal relationship.

When it comes to psychology, he argued that it could indeed someday become a science, but was unlikely to ever be an exact science. Predicting the behavior of human beings may be forever beyond our abilities, leaving us to limit ourselves to talking about **tendencies**.

His **utilitarianism** recognizes that happiness is not restricted to physical pleasures (or the avoidance of pain), that there may be different kinds or qualities of happiness. "It is better to be a human being dissatisfied than a pig satisfied; better to be Socrates dissatisfied than a fool satisfied." So, although we certainly begin as simple pleasure-seeking creatures, over time we can acquire far more humanistic motivations. Ultimately, this means that high moral values can be taught, and are not dependent on innate qualities of character.

When looking at social issues, J. S. Mill applies his expanded utilitarianism: Does a certain institution add to human welfare? Or are there better alternatives? He argues, for example, that women should be allowed to vote because women's self-interests can add balance to men's self-interests, and lead to a better society. He argues for personal freedom because it allows creative individuals to better contribute to society. On the other hand, he notes that free-market capitalism tends to result in inequity and poverty, and we would be better served by some form of socialism.

Thomas Brown (1778-1820) of the Scottish School puts the finishing touches to associationism: His laws of suggestion (i.e. association) were resemblance, contrast, and nearness in space and time, just like Aristotle's. He added a set of secondary laws -- duration, liveliness, frequency, and recency -- that strengthened suggestions. Then he considered as well the degrees of coexistence with other associations, constitutional differences of mind or temperament, differing circumstances of the moment, state of health or efficiency of the body, and prior habits. Finally, he understood association as an active process of an active, holistic mind.

Alexander Bain (1818-1903), a lifelong friend of John Stuart Mill, connected associationism with physiology. Accepting the law of contiguity, similarity, and frequency, he viewed them, as had Hartley, as neurological. He added the law of compound association, which says that most associations are among whole clusters of other associations. And he added the law of constructive association, which says that we can also actively, creatively, add to our associations ourselves.

One of Bain's basic principles is immortalized as the **Spencer-Bain principle**: The frequency or probability of a behavior rises if it is followed by a pleasurable event, and decreases if it is followed by a painful event. This is, of course, the same principle that the behaviorists would elaborate on a century later.

Bain has an even larger role in the history of psychology. First, he is often given the credit of having written two of the earliest textbooks in psychology -- **The Senses and the Intellect** (1855) and **Emotions and the Will** (1859), both of which went through many editions, and were used, for example, by William James. He also founded the first English-language psychological journal, called **Mind**, in January of 1876.

Hermann Ebbinghaus

The preceding people were essentially philosophers, not scientists. The first psychologist who made an effort to study association scientifically was Hermann Ebbinghaus.

Hermann Ebbinghaus was born on January 23, 1850, in Barmen, Germany. His father was a wealthy merchant, who encouraged his son to study. Hermann attended the University of Halle and the University of Berlin, and received his doctorate from the University of Bonn in 1873. While traveling through Europe, he came across a copy of Fechner's Elements of Psychophysics, which turned him on to psychology.

Ebbinghaus worked on his research at home in Berlin and published a book called **On Memory: An Investigation in Experimental Psychology** in 1885. Basically, his research involved the memorization of **nonsense syllables**, which consisted of a consonant, a vowel, and another consonant. He would select a dozen words, then attempt to master the list. He recorded the number of trials it took, as well as the effects of variations such as relearning old material, or the **meaningfulness** of the syllables. The results have been confirmed and are still valid today.

He also wrote the first article on intelligence testing of school children, and devised a sentence completion test that became a part of the Binet-Simon test. He also published textbooks on psychology in 1897 and 1902 that were very popular for many years. Hermann Ebbinghaus died in 1909, a clear precursor to today's cognitive movement.

The laws of association would continue to have a powerful influence in psychology. The Behaviorists, of course, focused on stimulus-stimulus and stimulus-response associations. The Gestalt psychologists elaborated on the various associations they termed the laws of Prägnanz. Among the cognitive psychologists, there are various theories of semantic association. And the physiological psychologists talk about the neurological bases for association. The idea appears to be here to stay. But then, as Greek and Medieval philosophers knew, association is just a simple description of a commonplace occurrance!

Psychophysics

Again and again, philosophers stated unequivocally that psychology could never be a science. The activities and the contents of the mind could not be measured, and therefore an objectivity such as that achieved in physics and chemistry was out of reach. Psychology would forever remain subjective!

This would finally change in the early 1800s. **Ernst Weber** (1795 to 1878) was born June 24 in Wittemburg, Germany, the third of 13 children! He received his doctorate from the University of Leipzig in 1815, in physiology. He began teaching there right after graduation, and continued until he retired in 1871.

His research was predominantly concerned with the senses of touch and kinesthesia (the experience of muscle position and movement). He was the first to clearly demonstrate the existence of kinesthesia, and showed that touch was actually a conglomerate sense composed of senses for pressure, temperature, and pain.

His chosen interests led him to certain techniques: First, there is the **two-point threshold**, which is a matter of measuring the smallest distance noticeable to touch at various parts of the body. For example, the tongue had the smallest threshold (1 mm), and the back had the largest (60 mm).

A second technique involved kinesthesia: **Just-noticeable difference** is the smallest difference in weight a person is capable of perceiving through holding two things. He discovered that the just-noticeable difference was a constant fraction of the weights involved. If you are holding a 40 pound weight in one hand, you will be able to recognize that a 41 pound weight in the other hand is in fact different. But if it were a 20 pound weight, you could detect a mere half pound difference! In other words, as regards weight, we could recognize a 1/40 difference, whatever the weights.

This is known as **Weber's Law**, and is the first such "law" relating a physical stimulus with a mental experience.

Gustav Fechner

Gustav Fechner was born April 1, 1801. His father, a village pastor, died early in Gustav's childhood, so he, with his mother and brother, went to live with their uncle. In 1817, at the age of 16, he went off to study medicine at the University of Leipzig (were Weber was teaching). He received his MD degree in 1822 at the age of 21.

But his interests moved to physics and math, so he made his living tutoring, translating, and occasionally lecturing. After writing a significant paper on electricity in 1831, he was invited to become a professor of physics at Leipzig. There, he became friends with a number of people, including Wilhelm Wundt, and his interests moved again, this time to psychology, especially vision.

In 1840, he had a nervous breakdown, and he had to resign his position due to severe depression. At his worst, he stayed in his rooms alone, avoiding light which hurt his eyes, and even painted his room black. While lying in bed one morning, October 22, 1850, he suddenly realized that it was indeed possible to connect the measurable physical world with the mental world, supposed to be inaccessible to scientific investigation! As his condition improved, he returned to writing and performing endless experiments, using mostly himself as a subject.

Like many people at the time, he found Spinoza's double-aspectism convincing and found in panpsychism something akin to a personal religion. Using the pseudonym **Dr. Mises**, he wrote a number of satires about the medicine and philosophy of his day. But he also used it to communicate, often in an amusing way, his spiritual perspective. As a panpsychist, he believed that all of nature was alive and capable of awareness of one degree or another. Even the planet earth itself, he believed, had a soul. He called this the **day-view**, and opposed it to the **night-view** of materialism.

Further, he felt that our lives come in three stages -- the fetal life, the ordinary life, and the life after death. When we die, our souls join with other souls as part of the supreme soul.

It was double-aspectism that led him to study (and name) **psychophysics**, which he defined as the study of the systematic relationships between physical events and mental events. In 1860, he topped his career by publishing the **Elements of Psychophysics**.

In this book, he introduced a mathematical expression of **Weber's Law**. The expression looked like this...

***R /R = k**

which means that the proportion of the minimum change in stimulus detectable (*R) to the strength of the stimulus (R) is a constant (k). (R is for the German Reiz, meaning stimulus.) Or...

S = k log R

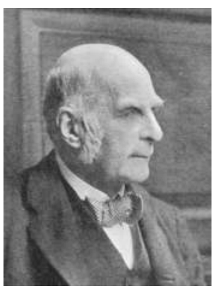
where S is the experienced sensation.

Fechner died November 28, 1887.

What Weber and Fechner showed that makes them far more significant than just Weber's Law is that psychological events are in fact tied to measurable physical events in a systematic way, which everyone had thought impossible. Psychology could be a science after all!

The second quantitative breakthrough would be the measurement of something far more complex, far more "psychological:" **intelligence**. We owe this to two great minds in particular: Sir Francis Galton in England and Alfred Binet in France.

Sir Francis Galton



Francis Galton was born February 16, 1822 near Birmingham, England. He was the youngest of 7 children, and first cousin of Charles Darwin. His father, a wealthy banker, insisted on educating Francis at home, especially considering that Francis could read at 2 and a half years old!

Later in childhood, he was sent off to boarding school, which he despised and criticized even in adulthood. At 16, he went to medical school at King's College at Oxford. He finished his degree at Cambridge in 1843, at 21.

His father died, leaving Galton a wealthy young aristocrat. He traveled extensively and became a member of the Royal Geographical Society, for which he developed maps of new territories and accounts of his adventures. He became president of that organization in 1856.

Galton had a penchant for measuring everything -- extending even to the behinds of women he encountered in his travels in Africa (something he had to do from a distance, of course, by means of triangulation). This interest in measurement led to his invention of the weathermap (including highs, lows, and front -- terms he introduced), and to suggesting the use of fingerprints to Scotland Yard.

His obsession eventually led to his efforts at measuring intelligence. In 1869, he published **Hereditary Genius: An Inquiry into its Laws and Consequences**, in which he demonstrates that the children of geniuses tend to be geniuses themselves.

In 1874, he produced **English Men of Science: Their Nature and Nurture**, based on long surveys passed out to thousands of established scientists. In this volume, he noted that, although the potential for high intelligence is still clearly inherited, it also needed to be nurtured to come to full fruition. In particular, the broad, liberal education provided by the Scottish school system proved far superior to the English school system he hated so much.

In 1883, he wrote **Inquiries into Human Faculty and its Development**. This would be the first time anyone compared identical and fraternal twins, a method now considered ideal when investigating nature vs nurture issues.

In 1888, he published **Co-Relations and Their Measurement, Chiefly from Anthropometric Data**. As the title suggests, it was Galton who invented correlation, as well as scatter plots and regression toward the mean. Later, Karl Pearson (1857-1936) would discover the mathematical formulation of correlation.

Sir Francis died in 1911, after an incredibly productive, if somewhat eccentric, life.

Alfred Binet

Born July 11, 1851 in Nice, France, Alfred was an only child. His mother, an artist, raised him by herself after a divorce from his father, a physician.

He started studying medicine, but decided to study psychology on his own -- being independently wealthy left him free to do what he pleased! He worked with the psychiatrist Charcot at La Salpetriere, where he studied hypnosis.

In 1891, he moved to Paris to study at the physiological-psychology lab at the Sorbonne, where he developed a variety of research interests, especially, of course, involving individual differences. In 1899, he and his graduate student, **Theodore Simon** (1873-1961) were commissioned by the French government to study retardation in the French schools, and to create a test to differentiate normal from retarded children.

After marriage, he began studying his own two daughters and testing them with Piaget-like tasks and other tests. This led to the publication of The Experimental Study of Intelligence in 1903.

In 1905, Binet and Simon came out with the **Binet-Simon Scale of Intelligence**, the first test permitting graduated, direct testing of intelligence. They expanded the test to normal children in 1908, and to adults in 1911.

Binet believed intelligence to be complex, with many factors, and not to be a simple, single entity. He didn't like the use of a single number as developed by **William Stern** in 1911 -- the intelligence quotient or IQ. He also believed that, though genetics may set upper limits on intelligence, most of us have plenty of room for improvement with the right kind of education.

He cautioned that his tests should be used with restraint: Even a child two years behind his age level may later prove to be brighter than most! He was afraid that IQ would prejudice teachers and parents, and that people would tend to view it as fixed and prematurely give up on kids who score low early on.

He suggested something he called **mental orthopedics**: Exercises in attention and thought that could help disadvantaged children "learn how to learn." He died in 1911, a man way ahead of his time, and wiser than most!

Binet's fears were well founded. For example, **Charles Spearman** (1863-1945) introduced the idea that "general intelligence" (g) was real, unitary, and inherited.

Worse were the antics of **Henry Goddard** (1866-1957). He translated the Binet Simon test into English. He studied a family in New Jersey he named the Kallikaks. Some were normal, but quite a few were "feebleminded" (Goddard's term). He traced their genealogy to support the heredity position. Because he believed that there was a close connection between feeblemindedness and criminality, he recommended that states institute programs of sterilization of the feebleminded. 20 states passed such laws.

Goddard also tested immigrants, at the request of the Immigration Service. His testers found 40 to 50% of immigrants feebleminded, and they were immediately deported. He also cited particular countries as being more feebleminded than others! Keep in mind that these immigrants rarely spoke much English and were tested during the grueling process of passing through the bureaucracy of Ellis Island after a long ocean voyage in miserable conditions!

Eugenics -- a term coined by Galton -- is the policy of intentionally breeding human beings according to some standard, and the sterilization of those that do not meet those standards. It became an institutionalized reality in 1907, when the Indiana legislature passed a law that made sterilization of "defectives" possible. A federal Eugenics Record Office was established in Cold Spring Harbor, and their lawyers designed law in 1914 that was promoted as models for the entire country.

Virginia adopted such a law in 1924. Emma Buck, her daughter Carrie and infant granddaughter Vivian, were judged to be feebleminded, and their case (Buck vs Bell) was taken before the Supreme Court. The Supreme Court, under Oliver Wendell Holmes, came down in support of the sterilization laws. Holmes stated:

> *"It is better for all the world, if instead of waiting to execute degenerate offspring for crime, or to let them starve for their imbecility, society can prevent those who are manifestly unfit from continuing their kind. The principle that sustains compulsory vaccination is broad enough to cover cutting the Fallopian tubes. Three generations of imbeciles are enough."*

Although scientists disputed the reasoning behind the sterilization laws, 33 states adopted them, and some 65,000 American citizens were sterilized. The Nazis based their eugenics laws on the American ones and sterilized 350,000. Eugenics gradually became unpopular as the horrors of Nazi Germany became public, and gradually ended in the 1940's. The Supreme Court has yet to reverse its opinion on the matter, however.

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CHAPTER OVERVIEW

2: Biology and Evolution

2.1: Darwin and Evolution

Charles Robert Darwin (1809-1882)

Charles Darwin was born in Shrewsbury, England, on February 2, 1809. His father was Robert Waring Darwin, a physician and son of the famous Erasmus Darwin, also a physician, as well as a respected writer and naturalist. His mother was Susannah Wedgwood Darwin. She died when Charles was eight.

Charles was educated in the local school, taught by Dr. Samuel Butler. In 1825, he went to Edinburgh to start studying medicine, but he soon realized that he did not have the stomach for it! So he switched to Cambridge, ostensibly to become a clergyman. He was actually more interested in entomology -- especially beetles -- and in hunting. He graduated from Christ's College in 1831.

It is said that even when he was a young man, he had a patient and open mind, spending many hours collecting specimens of one sort or another and pondering over new ideas. The idea of evolution was very much in the air in those times: It was increasingly clear to naturalists that species change and have been changing for many millennia. The question was, how did this happen?

One of his mentors, John Henslow, encouraged him to apply for the (unpaid!) position of naturalist on a surveying expedition on the now-famous vessel, the Beagle, under the command of Capt. Robert Fitz-Roy. Charles left England for the first time in his life on December 27, 1831. He wouldn't return until October 2, 1836!

Most of the ship's time was spent surveying the coasts of South America and nearby islands, but it would also visit various Pacific islands, New Zealand, and Australia. It was the Galapagos Islands that most impressed him. There he found that finches had evolved a variety of beaks -- each suited to a particular food source. Natural variation had somehow been selected to fit the ecological niches available on the tiny islands.

Upon returning, Darwin wrote several books based on his surveys on geology and the plant and animal species he had observed and collected. He also published his journal as **Journal of a Naturalist**. He notes that he was most impressed by the ways similar animals adapted to different ecologies.

From early on, Darwin recognized that selection was the principle men used so successfully when breeding animals. What he needed now was an idea as to how nature could perform that task without the benefit of intelligence!

In 1838, he read a book by Malthus called **Population**. Malthus introduced the idea that competition over limited resources would, in nature, keep populations stable. He also warned that human populations, when straining resources, would suffer as well!

On January 29, 1839, Darwin married Emma Wedgewood, a cousin. They lived in London for a few years, then settled in the village of Down, 15 miles outside London, where they lived the rest of their lives. Darwin began suffering from an illness he had probably contracted from an insect bite in the Andes many years before. Darwin's son, Francis, later could not say enough about his mother's dedication to his father's well-being. Without her, he would have been considerably less productive. They would go on to have two daughters and five sons.

Darwin wrote a sketch of his theory in 1842. In 1844, he wrote in a letter "At last gleams of light have come, and I am almost convinced (quite contrary to the opinion I started with) that species are not (it is like confessing a murder) immutable."

He was about half done with a full exposition of his ideas when he received an essay from A. R. Wallace, with a request for comments. The essay outlined a theory of natural selection! Wallace, too, had read Malthus, and in 1858, while sick from fever, had the whole idea come to him in one flash. Darwin, in his reluctance, had postponed revealing his ideas to the scientific public for 20 years!

Darwin forwarded the essay to his friend, Sir Charles Lyell of the Geological Society, as Wallace had requested. Lyell sent the essay on, with an essay by Darwin, for presentation at a scientific conference.

The point they jointly made was clear: Just like men can exaggerate one or another minor variation by selectively breeding dogs or cattle, so nature selects similar variations -- by only permitting the most successful variations to survive and reproduce in the struggle over limited resources. Although the changes would be slight and slow, the millennia would permit the obvious diversity of nature! Darwin named this **natural selection**.

In 1859, Darwin finally published his master work, on the **Origin of Species by Means of Natural Selection**. The book was an instant success. There was also, of course, a great deal of debate -- mostly concerning the contrast with traditional religious explanations of the natural world.

Natural selection was often confused with an earlier idea of the French naturalist Lamarck. He suggested that characteristics acquired during an animal's life were passed on to its offspring. The famous example is how the constant stretching of the neck over many generations explains the giraffe's unlikely physique. This theory -- **Lamarckianism** -- was natural selection's major competitor for decades to come!

In 1868, he published **The Variation of Animals and Plants under Domestication**. Then, in 1871, he came out with **The Descent of Man, and Selection in Relation to Sex**. This was really two books in one. The second part is about sexual selection. This is what accounts for, for example, the bright colors of many male birds: Both the males' coloring and the attraction to the coloring on the part of the females during courtship are selected for because these variations benefit the offspring.

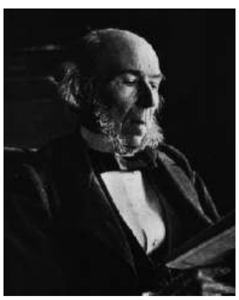
The **Descent of Man** portion of the book is a brief introduction to the idea that we, too, are the results of natural selection. This part would lead to a lot of heated arguments!

In 1872, **The Expression of Emotion** was published. This time, Darwin talks about the evolution of the signals that animals use to communicate -- and relates those signals to human emotional expression. This is the first step towards what we now call sociobiology (and evolutionary psychology).

In addition to these influential books, Darwin also enjoyed studying and writing about plants. In 1862, he wrote **Fertilization of Orchids**. In 1875, he came out with both **Climbing Plants** and **Insectivorous Plants**. In 1877, came **Different Forms of Flowers on Plants of the Same Species**. In 1880, he wrote, with his son Francis, **The Power of Movement in Plants**. And in 1881, he published the famous **Formation of Vegetable Mould through the Action of Worms**!

Charles Darwin died April 19, 1882. He was buried in Westminster Abbey. He was apparently a kind and gentle man, beloved by his family and friends alike. Outside of his voyage on the Beagle, he rarely left his home in Down. Reluctantly, he surrendered his religious beliefs and settled into an agnosticism that did not prevent him from participating with his parish in charitable works.

Herbert Spencer



Herbert Spencer was born April 27, 1820, in Derby, England. His father was a schoolmaster, and both his parents were "dissenters" (i.e. religious non-conformists). Spencer was clearly gifted and was mostly self-educated.

An excellent writer, he wrote articles on social issues for various magazines of the day and even became editor of **The Economist** for several years. In 1855, he published **The Principles of Psychology**. This became part of a series of books, which he called **The Synthetic Philosophy**, and included biology and sociology as well as psychology.

Originally believing in the inheritance of acquired characteristics (Lamarckianism), he became a follower of Darwin's theory. It was, in fact, Spencer who coined the phrase "survival of the fittest" But he also transformed Darwin's theory into a social theory that encouraged extreme individualism and laissez-faire economic policies, called **Social Darwinism.**

Basically, survival of the fittest was to apply to people competing against people as well, and he implied that it was something of a social duty to accept the fact that some would be rich and some poor -- and that the consequences of poverty should not be interfered with. Even whole societies -- such as England -- were engaged in a struggle for survival that did not allow for weakness of will or sentimentality.

Social Darwinism is not something Darwin would have approved of. It has in it the fallacy of false analogy: Human society is not a neat parallel to the non-human biological world. Unfortunately, Social Darwinism seems to be here to stay, and can be found within Fascist, Conservative, and Libertarian political agendas, and in personal philosophies such as that of Ayn Rand.

Spencer is, nevertheless, considered one of the great productive thinkers of his day. He died Dec. 8, 1903, in Brighton, Sussex.

(A major resource for Darwin's biography was the 11th edition (1910/1911) of the **Encyclopedia Britannica**, available online here)

2.2: Eugenics and its Evolution in the History of Western Psychology

Read this!

Eugenics and its Evolution in the History of Western Psychology by Oksana Yakushko

3: Freud and Psychoanalysis

Precursors of Psychoanalysis

It often surprises students that **psychiatry** - meaning the doctoring of the mind - was not invented by Sigmund Freud. Psychoanalysis - a particular (and very significant) brand of psychiatry - was his baby. Psychiatrists existed before Freud, and most psychiatrists today are not Freudian.

The term psychiatry was coined by the German physician **Johann Reil**[1] in 1808, and would slowly replace the older term "alienist." The new respect signalled by the new name was based on some significant improvements in the care of the mentally ill in the second half of the 1700's.

There are three people I would like to pay my respects to as important precursors to psychoanalysis: **Franz Anton Mesmer**, who discovered hypnotism; **Philippe Pinel**, who changed the way we thought of and treated the mentally ill; and **Jean-Martin Charcot**, who is often considered the father of neurology.

Franz Anton Mesmer

Franz Anton Mesmer was born May 23, 1734 in Iznang, Germany, near Lake Constance. He received his MD from the University of Vienna in 1766. His dissertation concerned the idea that the planets influenced the health of those of us on earth. He suggested that their gravitational forces could change the distribution of our animal spirits. Later, he changed his theory to emphasize magnetism rather than gravity -- hence the term "**animal magnetism**." It would soon, however, come to be known as **mesmerism**.

He was, in fact, able to put people into trance states, even convulsions, by waving magnetized bars over them. His dramatic performances were quite popular for a while, although he believed that anyone could achieve the same results. In point of fact, some of his patients did in fact get relief from their symptoms -- a point that would later be investigated by others.

When accused of fraud by other physicians in Vienna, he went to Paris. In 1784, the King of France, Louis XVI, appointed a commission including Benjamin Franklin to look into Mesmer and his practices. They concluded that his results were due to nothing more than **suggestion**.

Despite condemnation by many of the educated elite, mesmerism became a popular fad in the salons of Europe. In order to serve the many poor people who came to him for help, he designed a sort of bathtub in which they could sit while holding the magnetic rods themselves. He eventually created an organization to train other mesmerists.

Mesmer died March 5, 1815 in Meersburg, also near Lake Constance, Germany.

An English physician, **James Braid** (1795-1860), a much more careful researcher of Mesmer's phenomenon, termed it **hypnotism**. Disassociated from Mesmer, hypnotism would go on to have a long, if controversial, life into the twentieth century.

Philippe Pinel

Philippe Pinel was born on April 20, 1745, in the small town of Saint André. His father was both a barber and a surgeon, a common combination in those days, as both vocations required a steady hand with the razor. His mother was also from a long line of physicians.

Philippe began his studies more interested in literature -- especially Jean-Jacques Rousseau -- than in medicine. But, after a few years studying theology, he began the study of medicine, and he recieved his MD from the University at Toulouse in 1773.

Pinel moved to Montpellier in 1774 where he tutored wealthy students in anatomy and mathematics. He was admited into the Montpellier Société Royale des Sciences after presenting two papers on the use of mathematics in anatomical studies. He moved to Paris in 1778, where he came into contact with a number of the renowned scientists and philosophers of the day (including Ben Franklin), as well as becoming familiar with the radical new ideas of John Locke and the French sensationalists. Although he could not practice in Paris, he became a well respected medical writer, particularly known for his careful and exhaustive **case studies**.

A turning point in Pinel's life came in 1785, when a friend of his developed a mental illness ending in his death. He became devoted to the study of mental illness, and became the head of the Paris asylum for insane men at Bicêtre in 1792. In that year, he also married Jeanne Vincent, with whom he had three sons.

It was at Bicêtre that he made his place in history: Prior to his coming to Bicêtre, the men were kept in chains, treated abominably, and put on daily display to the public as curiosities. In 1793, Pinel instituted a new program of human care, which he referred to as **moral therapy**. The men were given clean, comfortable accommodations, and were instructed in simple but productive work.

In 1795, he was appointed the head physician at the world famous hospital at Salpêtrière. Here, too, he provided his enlightened treatment conditions to the mentally ill. In that same year, he was made professor of medical pathology at Paris. In 1801, Phillipe Pinel introduced the first textbook on moral therapy to the world.

Pinel is also remembered for dismissing the demonic possession theory of mental illness for once and for all, and for eliminating treatments such as bleeding from his hospital. He also introduced other novelties to his hospital, such as vaccinations and the use of the stethoscope. He was a physician to Napoleon and was made a knight of the Legion d'Honneur in 1804. He died in Paris on October 25, 1826.

Pinel's innovations were soon imitated in other countries, by such notables as **William Tuke** in England, **Vincenzo Chiarugi** in Florence, and **Dorothea Dix** in the U.S.

Jean-Martin Charcot

Jean-Martin Charcot was born in Paris on November 29, 1825. He received his MD at the University of Paris in 1853. In 1860 he became a professor at his alma mater. Two years later, he began to work at Salpêtrière Hospital as well. In 1882, he opened a neurological clinic at Salpêtrière Hospital. It, and he, became known throughout Europe, and students came from everywhere to study the new field. Among them were Alfred Binet and a young Sigmund Freud.

Charcot is well known in medical circles for his studies of the neurology of motor disorders, resulting diseases, aneurysms, and localization of brain functions. He is considered the father of modern neurology as well as the person who first diagnosed Multiple Sclerosis.

In psychology, he is best known for his use of hypnosis to successfully treat women suffering from the psychological disorder then known as **hysteria**. Now called **conversion disorder**, hysteria involved a loss of some physiological function such as vision, speech, tactile sensations, movement, etc., that was nonetheless not based in actual neurological damage.

Charcot believed that hysteria was due to a congenitally weak nervous system, combined with the effects of some traumatic experience. Hypnotizing these patients brought on a state similar to hysteria itself. He found that, in some cases, the symptoms would actually lessen after hypnosis -- although he was only interested in studying hysteria, not in curing it! Others would later use hypnosis as a part of curing the problem.

Charcot died in Morvan, France, on August 16, 1893..

The Unconscious

Before we turn to the really big names, let's take a peek at the concept of the unconscious, so strongly associated with psychoanalysis. Most historians agree that the first mention of such a concept was Leibniz's discussion of "**petites perceptions**" or little perceptions. By this he meant certain very low-level stimuli that could enter the mind without the person's awareness - what today we would call subliminal messages. The reality of such things is very much in doubt.

Johann Friedrich Herbart (1776-1841) was the author of a textbook on psychology, published in 1816. But, following Kant, he did not believe psychology could ever be a science. He took the concepts of the associationists and blended them with the dynamics of Leibniz's monads. Ideas had an energy of their own, he said, and could actually force themselves on the person's conscious mind by exceeding a certain threshold. When ideas were incompatible, one or the other would be **repressed**, he said - meaning forced below the threshold into the unconscious. This should remind you of Freud's ideas - except that Herbart had them nearly a century earlier!

Schopenhauer is often seen as the originator of the unconscious, and he spoke at great lengths about instincts and the irrational nature of man, and freely made use of words like repression, resistance, and sublimation! **Nietzsche** also spoke of the unconscious: One of his most famous statements is "My memory says I did it. My pride says I could not have done that. In the end, my memory yields."

Johann Friedrich Herbart

One more pre-Freudian should be mentioned: **Karl Eduard von Hartmann** (1842-1906). He blended the ideas of Schopenhauer with Jewish mysticism (the kaballah) and wrote **Philosophy of the Unconscious** in 1869, just in time to influence a young neurologist named Sigmund Freud.

The reader should understand that there are many theorists with little or no use for the concept of the unconscious. Brentano, forefather of phenomenology and existentialism, did not believe in it. Neither did William James. Neither did the Gestalt psychologists. Memories, for example, can be understood as stored in some physical state, perhaps as traces in the brain. When activated, we remember. But they aren't in the mind - conscious or unconscious - until so activated.

In addition to the concept of the unconscious, another early landmark of psychiatry was the introduction of careful diagnosis of mental illness, beginning with **Emil Kraepelin's** work (1856-1926). The first differentiated classification was of what he labelled **dementia praecox**, which meant the insanity of adolescence. Kraepelin also invented the terms **neurosis** and **psychosis**, and named **Alzheimer's disease** after **Alois Alzheimer**, who first described it. I should also mention **Eugen Bleuler**, who coined the term **schizophrenia** to replace dementia praecox in 1911.

Now, on to Freud....

Sigmund Freud

Freud's story, like most people's stories, begins with others. In his case those others were his mentor and friend, Dr. Joseph Breuer, and Breuer's patient, called Anna O.

Anna O. was Joseph Breuer's patient from 1880 through 1882. Twenty one years old, Anna spent most of her time nursing her ailing father. She developed a bad cough that proved to have no physical basis. She developed some speech difficulties, then became mute, and then began speaking only in English, rather than her usual German.

When her father died she began to refuse food, and developed an unusual set of problems. She lost the feeling in her hands and feet, developed some paralysis, and began to have involuntary spasms. She also had visual hallucinations and tunnel vision. But when specialists were consulted, no physical causes for these problems could be found.

If all this weren't enough, she had fairy-tale fantasies, dramatic mood swings, and made several suicide attempts. Breuer's diagnosis was that she was suffering from what was then called **hysteria** (now called conversion disorder), which meant she had symptoms that appeared to be physical, but were not.

In the evenings, Anna would sink into states of what Breuer called "spontaneous hypnosis," or what Anna herself called "clouds." Breuer found that, during these trance-like states, she could explain her day-time fantasies and other experiences, and she felt better afterwards. Anna called these episodes "chimney sweeping" and "the talking cure."

Sometimes during "chimney sweeping," some emotional event was recalled that gave meaning to some particular symptom. The first example came soon after she had refused to drink for a while: She recalled seeing a woman drink from a glass that a dog had just drunk from. While recalling this, she experienced strong feelings of disgust...and then had a drink of water! In other words, her symptom -- an avoidance of water -- disappeared as soon as she remembered its root event, and experienced the strong emotion that would be appropriate to that event. Breuer called this catharsis, from the Greek word for cleansing.

It was eleven years later that Breuer and his assistant, Sigmund Freud, wrote a book on hysteria. In it they explained their theory: Every hysteria is the result of a traumatic experience, one that cannot be integrated into the person's understanding of the world. The emotions appropriate to the trauma are not expressed in any direct fashion, but do not simply evaporate: They express themselves in behaviors that in a weak, vague way offer a response to the trauma. These symptoms are, in other words, meaningful. When the client can be made aware of the meanings of his or her symptoms (through hypnosis, for example) then the unexpressed emotions are released and so no longer need to express themselves as symptoms. It is analogous to lancing a boil or draining an infection.

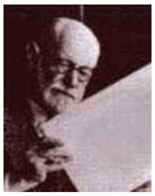
In this way, Anna got rid of symptom after symptom. But it must be noted that she needed Breuer to do this: Whenever she was in one of her hypnotic states, she had to feel his hands to make sure it was him before talking! And sadly, new problems continued to arise.

According to Freud, Breuer recognized that she had fallen in love with him, and that he was falling in love with her. Plus, she was telling everyone she was pregnant with his child. You might say she wanted it so badly that her mind told her body it was true, and she developed an hysterical pregnancy. Breuer, a married man in a Victorian era, abruptly ended their sessions together, and lost all interest in hysteria. Please understand that recent research suggests that many of these events, including the hysterical pregnancy and Breuer's quick retreat, were probably Freud's "elaborations" on reality!

It was Freud who would later add what Breuer did not acknowledge publicly - that secret sexual desires lay at the bottom of all these hysterical neuroses.

To finish her story, Anna spent time in a sanatorium. Later, she became a well-respected and active figure - the first social worker in Germany - under her true name, Bertha Pappenheim. She died in 1936. She will be remembered, not only for her own accomplishments, but as the inspiration for the most influential personality theory we have ever had.

Biography



Sigmund Freud was born May 6, 1856, in a small town - Freiberg - in Moravia. His father was a wool merchant with a keen mind and a good sense of humor. His mother was a lively woman, her husband's second wife and 20 years younger. She was 21 years old when she gave birth to her first son, her darling, Sigmund. Sigmund had two older half-brothers and six younger siblings. When he was four or five - he wasn't sure - the family moved to Vienna, where he lived most of his life.

A brilliant child, always at the head of his class, he went to medical school, one of the few viable options for a bright Jewish boy in Vienna those days. There, he became involved in research under the direction of a physiology professor named Ernst Brücke. Brücke believed in what was then a popular, if radical, notion, which we now call reductionism: "No other forces than the common physical-chemical ones are active within the organism." Freud would spend many years trying to "reduce" personality to neurology, a cause he later gave up on.

Freud was very good at his research, concentrating on neurophysiology, even inventing a special cell-staining technique. But only a limited number of positions were available, and there were others ahead of him. Brücke helped him to get a grant to study, first with the great psychiatrist Charcot in Paris, then with his rival Bernheim in Nancy. Both these gentlemen were investigating the use of hypnosis with hysterics.

After spending a short time as a resident in neurology and director of a children's ward in Berlin, he came back to Vienna, married his patient fiancée Martha Bernays, and set up a practice in neuropsychiatry, with the help of Joseph Breuer.

It is true that Freud experimented with cocaine, encouraged his fiancée and friends to try it, and wrote about its great restorative powers. He later came to regret his enthusiasm for the drug and withdrew his endorsement.

Freud's books and lectures brought him both fame and ostracism from the mainstream of the medical community. He drew around him a number of very bright sympathizers who became the core of the psychoanalytic movement. Unfortunately, Freud had a penchant for rejecting people who did not totally agree with him. Some separated from him on friendly terms; others did not, and went on to found competing schools of thought.

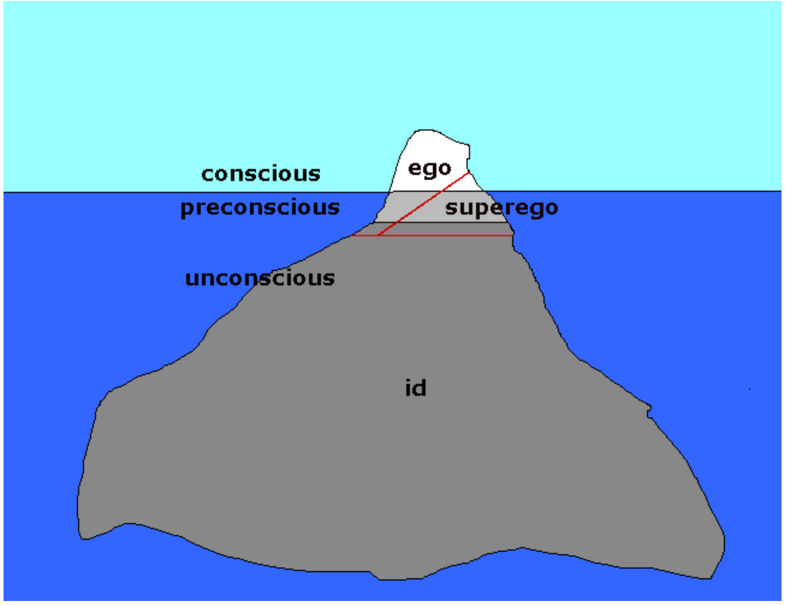
Freud emigrated to England just before World War II when Vienna became an increasing dangerous place for Jews, especially ones as famous as Freud. Not long afterward, he died of the cancer of the mouth and jaw that he had suffered from for the last 20 years of his life.

Theory

Freud didn't exactly invent the idea of the conscious versus unconscious mind, but he certainly was responsible for making it popular. The **conscious mind** is what you are aware of at any particular moment, your present perceptions, memories, thoughts, fantasies, feelings, what have you. Working closely with the conscious mind is what Freud called the **preconscious**, what we might today call "available memory:" anything that can easily be made conscious, the memories you are not at the moment thinking about but can readily bring to mind. Now no-one has a problem with these two layers of mind. But Freud suggested that these are the smallest parts!

The largest part by far is the **unconscious**. It includes all the things that are not easily available to awareness, including many things that have their origins there, such as our drives or instincts, and things that are put there because we can't bear to look at them, such as the memories and emotions associated with trauma.

According to Freud, the unconscious is the source of our motivations, whether they be simple desires for food or sex, neurotic compulsions, or the motives of an artist or scientist. And yet, we are often driven to deny or resist becoming conscious of these motives, and they are often available to us only in disguised form. We will come back to this.



The id, the ego, and the superego

Freudian psychological reality begins with the world, full of objects. Among them is a very special object, the organism. The organism is special in that it acts to survive and reproduce, and it is guided toward those ends by its needs -- hunger, thirst, the avoidance of pain, and sex.

A part - a very important part - of the organism is the nervous system, which has as one of its characteristics a sensitivity to the organism's needs. At birth, that nervous system is little more than that of any other animal, an "it" or **id**. The nervous system, as id, translates the organism's needs into motivational forces called, in German, **Trieben**, which has been translated as **instincts** or **drives**. Freud also called them **wishes**. This translation from need to wish is called the **primary process**.

The id works in keeping with the **pleasure principle**, which can be understood as a demand to take care of needs immediately. Just picture the hungry infant, screaming itself blue. It doesn't "know" what it wants in any adult sense; it just knows that it wants it and it wants it now. The infant, in the Freudian view, is pure, or nearly pure id. And the id is nothing if not the psychic representative of biology.

Unfortunately, although a wish for food, such as the image of a juicy steak, might be enough to satisfy the id, it isn't enough to satisfy the organism. The need only gets stronger, and the wishes just keep coming. You may have noticed that, when you haven't satisfied some need, such as the need for food, it begins to demand more and more of your attention, until there comes a point where you can't think of anything else. This is the wish or drive breaking into consciousness.

Luckily for the organism, there is that small portion of the mind we discussed before, the conscious, that is hooked up to the world through the senses. Around this little bit of consciousness, during the first year of a child's life, some of the "it" becomes "I," some of the id becomes **ego**. The ego relates the organism to reality by means of its consciousness, and it searches for objects to satisfy the wishes that id creates to represent the organisms needs. This problem-solving activity is called the **secondary process**.

The ego, unlike the id, functions according to the **reality principle**, which says "take care of a need as soon as an appropriate object is found." It represents reality and, to a considerable extent, reason.

However, as the ego struggles to keep the id (and, ultimately, the organism) happy, it meets with obstacles in the world. It occasionally meets with objects that actually assist it in attaining its goals. And it keeps a record of these obstacles and aides. In particular, it keeps track of the rewards and punishments meted out by two of the most influential objects in the world of the child - mom and dad. This record of things to avoid and strategies to take becomes the **superego**. It is not completed until about seven years of age. In some people, it never is completed.

There are two aspects to the superego: One is the **conscience**, which is an internalization of punishments and warnings. The other is called the **ego ideal**. It derives from rewards and positive models presented to the child. The conscience and ego ideal communicate their requirements to the ego with feelings like pride, shame, and guilt.

It is as if we acquired, in childhood, a new set of needs and accompanying wishes, this time of social rather than biological origins. Unfortunately, these new wishes can easily conflict with the ones from the id. You see, the superego represents society, and society often wants nothing better than to have you never satisfy your needs at all!

The stages

Freud noted that, at different times in our lives, different parts of our skin give us greatest pleasure. Later theorists would call these areas **erogenous zones**. It appeared to Freud that the infant found its greatest pleasure in sucking, especially at the breast. In fact, babies have a penchant for bringing nearly everything in their environment into contact with their mouths. A bit later in life, the child focuses on the anal pleasures of holding it in and letting go. By three or four, the child may have discovered the pleasure of touching or rubbing against his or her genitalia. Only later, in our sexual maturity, do we find our greatest pleasure in sexual intercourse. In these observations, Freud had the makings of a psychosexual stage theory.

The **oral stage** lasts from birth to about 18 months. The focus of pleasure is, of course, the mouth. Sucking and biting are favorite activities.

The **anal stage** lasts from about 18 months to three or four years old. The focus of pleasure is the anus. Holding it in and letting it go are greatly enjoyed.

The **phallic stage** lasts from three or four to five, six, or seven years old. The focus of pleasure is the genitalia. Masturbation is common.

The **latent stage** lasts from five, six, or seven to puberty, that is, somewhere around 12 years old. During this stage, Freud believed that the sexual impulse was suppressed in the service of learning. I must note that, while most children seem to be fairly calm, sexually, during their grammar school years, perhaps up to a quarter of them are quite busy masturbating and playing "doctor." In Freud's repressive era, these children were, at least, quieter than their modern counterparts.

The **genital stage** begins at puberty, and represents the resurgence of the sex drive in adolescence, and the more specific focusing of pleasure in sexual intercourse. Freud felt that masturbation, oral sex, homosexuality, and many other things we find acceptable in adulthood today, were immature.

This is a true stage theory, meaning that Freudians believe that we all go through these stages, in this order, and pretty close to these ages.

The Oedipal crisis

Each stage has certain difficult tasks associated with it where problems are more likely to arise. For the oral stage, this is weaning. For the anal stage, it's potty training. For the phallic stage, it is the Oedipal crisis, named after the ancient Greek story of king Oedipus, who inadvertently killed his father and married his mother.

Here's how the Oedipal crisis works: The first love-object for all of us is our mother. We want her attention, we want her affection, we want her caresses, we want her, in a broadly sexual way. The young boy, however, has a rival for his mother's charms: his father! His father is bigger, stronger, smarter, and he gets to sleep with mother, while junior pines away in his lonely little bed. Dad is the enemy.

About the time the little boy recognizes this archetypal situation, he has become aware of some of the more subtle differences between boys and girls, the ones other than hair length and clothing styles. From his naive perspective, the difference is that he has a penis, and girls do not. At this point in life, it seems to the child that having something is infinitely better than not having something, and so he is pleased with this state of affairs.

But the question arises: where is the girl's penis? Perhaps she has lost it somehow. Perhaps it was cut off. Perhaps this could happen to him! This is the beginning of **castration anxiety**, a slight misnomer for the fear of losing one's penis.

To return to the story, the boy, recognizing his father's superiority and fearing for his penis, engages some of his ego defenses: He displaces his sexual impulses from his mother to girls and, later, women; And he identifies with the aggressor, dad, and attempts to become more and more like him, that is to say, a man. After a few years of latency, he enters adolescence and the world of mature heterosexuality.

The girl also begins her life in love with her mother, so we have the problem of getting her to switch her affections to her father before the Oedipal process can take place. Freud accomplishes this with the idea of **penis envy**: The young girl, too, has noticed the difference between boys and girls and feels that she, somehow, doesn't measure up. She would like to have one, too, and all the power associated with it. At very least, she would like a penis substitute, such as a baby. As every child knows, you need a father as well as a mother to have a baby, so the young girl sets her sights on dad.

Dad, of course, is already taken. The young girl displaces from him to boys and men, and identifies with mom, the woman who got the man she really wanted. Note that one thing is missing here: The girl does not suffer from the powerful motivation of castration anxiety, since she cannot lose what she doesn't have. Freud felt that the lack of this great fear accounts for fact (as he saw it) that women were both less firmly heterosexual than men and somewhat less morally-inclined.

Before you get too upset by this less-than-flattering account of women's sexuality, rest assured that many people have responded to it. I will discuss it in the discussion section.

Therapy

Freud's therapy has been more influential than any other, and more influential than any other part of his theory. Here are some of the major points:

Relaxed atmosphere. The client must feel free to express anything. The therapy situation is in fact a unique social situation, one where you do not have to be afraid of social judgment or ostracism. In fact, in Freudian therapy, the therapist practically disappears. Add to that the physically relaxing couch, dim lights, sound-proof walls, and the stage is set.

Free association. The client may talk about anything at all. The theory is that, with relaxation, the unconscious conflicts will inevitably drift to the fore. It isn't far off to see a similarity between Freudian therapy and dreaming! However, in therapy, there is the therapist, who is trained to recognize certain clues to problems and their solutions that the client would overlook.

Resistance. One of these clues is resistance. When a client tries to change the topic, draws a complete blank, falls asleep, comes in late, or skips an appointment altogether, the therapist says "aha!" These resistances suggest that the client is nearing something in his free associations that he - unconsciously, of course - finds threatening.

Dream analysis. In sleep, we are somewhat less resistant to our unconscious and we will allow a few things, in symbolic form, of course, to come to awareness. These wishes from the id provide the therapist and client with more clues. Many forms of therapy make use of the client's dreams, but Freudian interpretation is distinct in the tendency to find sexual meanings.

Parapraxes. A parapraxis is a slip of the tongue, often called a Freudian slip. Freud felt that they were also clues to unconscious conflicts. Freud was also interested in the jokes his clients told. In fact, Freud felt that almost everything meant something almost all the time - dialing a wrong number, making a wrong turn, misspelling a word, were serious objects of study for Freud. However, he himself noted, in response to a student who asked what his cigar might be a symbol for, that "sometimes a cigar is just a cigar." Or is it?

Other Freudians became interested in **projective tests**, such as the famous Rorschach or inkblot tests. The theory behind these tests is that, when the stimulus is vague, the client fills it with his or her own unconscious themes. Again, these could provide the therapist with clues.

Transference, catharsis, and insight

Transference occurs when a client projects feelings toward the therapist that more legitimately belong with certain important others. Freud felt that transference was necessary in therapy in order to bring the repressed emotions that have been plaguing the client for so long, to the surface. You can't feel really angry, for example, without a real person to be angry at. The relationship between the client and the therapist, contrary to popular images, is very close in Freudian therapy, although it is understood that it can't get out of hand.

Catharsis is the sudden and dramatic outpouring of emotion that occurs when the trauma is resurrected. The box of tissues on the end table is not there for decoration.

Insight is being aware of the source of the emotion, of the original traumatic event. The major portion of the therapy is completed when catharsis and insight are experienced. What should have happened many years ago - because you were too little to deal with it, or under too many conflicting pressures - has now happened, and you are on your way to becoming a happier person.

Freud said that the goal of therapy is simply "to make the unconscious conscious."

Discussion

The only thing more common than a blind admiration for Freud seems to be an equally blind hatred for him. Certainly, the proper attitude lies somewhere in between. Let's start by exploring some of the apparent flaws in his theory.

The least popular part of Freud's theory is the Oedipal complex and the associated ideas of castration anxiety and penis envy. What is the reality behind these concepts? It is true that some children are very attached to their opposite-sex parent, and very competitive with their same-sex parent. It is true that some boys worry about the differences between boys and girls, and fear that someone may cut their penis off. It is true that some girls likewise are concerned, and wish they had a penis. And it is true that some of these children retain these affections, fears, and aspirations into adulthood.

Most personality theorists, however, consider these examples aberrations rather than universals, exceptions rather than rules. They occur in families that aren't working as well as they should, where parents are unhappy with each other, use their children against each other. They occur in families where parents literally denigrate girls for their supposed lack, and talk about cutting off the penises of unruly boys. They occur especially in neighborhoods where correct information on even the simplest sexual facts is not forthcoming, and children learn mistaken ideas from other children.

If we view the Oedipal crisis, castration anxiety, and penis envy in a more metaphoric and less literal fashion, they are useful concepts: We do love our mothers and fathers as well as compete with them. Children probably do learn the standard heterosexual behavior patterns by imitating the same-sex parent and practicing on the opposite-sex parent. In a male-dominated society, having a penis - being male - is better than not, and losing one's status as a male is scary. And wanting the privileges of the male, rather than the male organ, is a reasonable thing to expect in a girl with aspirations. But Freud did not mean for us to take these concepts metaphorically. Some of his followers, however, did.

Sexuality

A more general criticism of Freud's theory is its emphasis on sexuality. Everything, both good and bad, seems to stem from the expression or repression of the sex drive. Many people question that, and wonder if there are any other forces at work. Freud himself later added the death instinct, but that proved to be another one of his less popular ideas.

First let me point out that, in fact, a great deal of our activities are in some fashion motivated by sex. If you take a good hard look at our modern society, you will find that most advertising uses sexual images, that movies and television programs often don't sell well if they don't include some titillation, that the fashion industry is based on a continual game of sexual hide-and-seek, and that we all spend a considerable portion of every day playing "the mating game." Yet we still don't feel that all life is sexual.

But Freud's emphasis on sexuality was not based on the great amount of obvious sexuality in his society -- it was based on the intense avoidance of sexuality, especially among the middle and upper classes, and most especially among women. What we too easily forget is that the world has changed rather dramatically over the last hundred years. We forget that doctors and ministers recommended strong punishment for masturbation, that "leg" was a dirty word, that a woman who felt sexual desire was automatically considered a potential prostitute, that a bride was often taken completely by surprise by the events of the wedding night, and could well faint at the thought.

It is to Freud's credit that he managed to rise above his culture's sexual attitudes. Even his mentor Breuer and the brilliant Charcot couldn't fully acknowledge the sexual nature of their clients' problems. Freud's mistake was more a matter of generalizing too far, and not taking cultural change into account. It is ironic that much of the cultural change in sexual attitudes was in fact due to Freud's work!

The unconscious

One last concept that is often criticized is the unconscious. It is not disputed that something like the unconscious accounts for some of our behavior, but rather how much and the exact nature of the beast.

Behaviorists, humanists, and existentialists all believe that (a) the motivations and problems that can be attributed to the unconscious are much fewer than Freud thought, and (b) the unconscious is not the great churning cauldron of activity he made it out to be. Most psychologists today see the unconscious as whatever we don't need or don't want to see. Some theorists don't use the concept at all.

On the other hand, at least one theorist, Carl Jung, proposed an unconscious that makes Freud's look puny! But we will leave all these views for the appropriate chapters.

Positive aspects

People have the unfortunate tendency to "throw the baby out with the bath water." If they don't agree with ideas a, b, and c, they figure x, y, and z must be wrong as well. But Freud had quite a few good ideas, so good that they have been incorporated into many other theories, to the point where we forget to give him credit.

First, Freud made us aware of two powerful forces and their demands on us. Back when everyone believed people were basically rational, he showed how much of our behavior was based on biology. When everyone conceived of people as individually responsible for their actions, he showed the impact of society. When everyone thought of male and female as roles determined by nature or God, he showed how much they depended on family dynamics. The id and the superego - the psychic manifestations of biology and society - will always be with us in some form or another.

Second is the basic theory, going back to Breuer, of certain neurotic symptoms as caused by psychological traumas. Although most theorists no longer believe that all neurosis can be so explained, or that it is necessary to relive the trauma to get better, it has become a common understanding that a childhood full of neglect, abuse, and tragedy tends to lead to an unhappy adult.

Third is the idea of ego defenses. Even if you are uncomfortable with Freud's idea of the unconscious, it is clear that we engage in little manipulations of reality and our memories of that reality to suit our own needs, especially when those needs are strong. I would recommend that you learn to recognize these defenses: You will find that having names for them will help you to notice them in yourself and others!

Finally, the basic form of therapy has been largely set by Freud. Except for some behaviorist therapies, most therapy is still "the talking cure," and still involves a physically and socially relaxed atmosphere. And, even if other theorists do not care for the idea of transference, the highly personal nature of the therapeutic relationship is generally accepted as important to success.

Some of Freud's ideas are clearly tied to his culture and era. Other ideas are not easily testable. Some may even be a matter of Freud's own personality and experiences. But Freud was an excellent observer of the human condition, and enough of what he said has relevance today that he will be a part of personality textbooks for years to come. Even when theorists come up with dramatically different ideas about how we work, they compare their ideas with Freud's.

Carl Jung

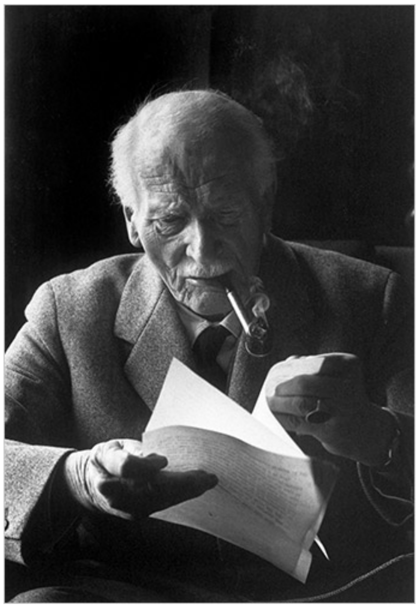
Freud said that the goal of therapy was to make the unconscious conscious. He certainly made that the goal of his work as a theorist. And yet he makes the unconscious sound very unpleasant, to say the least: It is a cauldron of seething desires, a bottomless pit of perverse and incestuous cravings, a burial ground for frightening experiences which nevertheless come back to haunt us. Frankly, it doesn't sound like anything I'd like to make conscious!

A younger colleague of his, Carl Jung, was to make the exploration of this "inner space" his life's work. He went equipped with a background in Freudian theory, of course, and with an apparently inexhaustible knowledge of mythology, religion, and philosophy. Jung was especially knowledgeable in the symbolism of complex mystical traditions such as Gnosticism, Alchemy, Kabbalah, and similar traditions in Hinduism and Buddhism. If anyone could make sense of the unconscious and its habit of revealing itself only in symbolic form, it would be Carl Jung.

He had, in addition, a capacity for very lucid dreaming and occasional visions. In the fall of 1913, he had a vision of a "monstrous flood" engulfing most of Europe and lapping at the mountains of his native Switzerland. He saw thousands of people drowning and civilization crumbling. Then, the waters turned into blood. This vision was followed, in the next few weeks, by dreams of eternal winters and rivers of blood. He was afraid that he was becoming psychotic.

But on August 1 of that year, World War I began. Jung felt that there had been a connection, somehow, between himself as an individual and humanity in general that could not be explained away. From then until 1928, he was to go through a rather painful process of self-exploration that formed the basis of all of his later theorizing.

He carefully recorded his dreams, fantasies, and visions, and drew, painted, and sculpted them as well. He found that his experiences tended to form themselves into persons, beginning with a wise old man and his companion, a little girl. The wise old man evolved, over a number of dreams, into a sort of spiritual guru. The little girl became "anima," the feminine soul, who served as his main medium of communication with the deeper aspects of his unconscious.



A leathery brown dwarf would show up guarding the entrance to the unconscious. He was "the shadow," a primitive companion for Jung's ego. Jung dreamt that he and the dwarf killed a beautiful blond youth, whom he called Siegfried. For Jung, this represented a warning about the dangers of the worship of glory and heroism which would soon cause so much sorrow all over Europe - and a warning about the dangers of some of his own tendencies towards hero-worship, of Sigmund Freud!

Jung dreamt a great deal about the dead, the land of the dead, and the rising of the dead. These represented the unconscious itself - not the "little" personal unconscious that Freud made such a big deal out of, but a new **collective unconscious** of humanity itself, an unconscious that could contain all the dead, not just our personal ghosts. Jung began to see the mentally ill as people who are haunted by these ghosts, in an age where no-one is supposed to even believe in them. If we could only recapture our mythologies, we would understand these ghosts, become comfortable with the dead, and heal our mental illnesses.

Critics have suggested that Jung was, very simply, ill himself when all this happened. But Jung felt that, if you want to understand the jungle, you can't be content just to sail back and forth near the shore. You've got to get into it, no matter how strange and frightening it might seem.

Biography

Carl Gustav Jung was born July 26, 1875, in the small Swiss village of Kessewil. His father was Paul Jung, a country parson, and his mother was Emilie Preiswerk Jung. He was surrounded by a fairly well educated extended family, including quite a few clergymen and some eccentrics as well.

The elder Jung started Carl on Latin when he was six years old, beginning a long interest in language and literature -- especially ancient literature. Besides most modern western European languages, Jung could read several ancient ones, including Sanskrit, the language of the original Hindu holy books.

Carl was a rather solitary adolescent, who didn't care much for school, and especially couldn't take competition. He went to boarding school in Basel, Switzerland, where he found himself the object of a lot of jealous harassment. He began to use sickness as an excuse, developing an embarrassing tendency to faint under pressure.

Although his first career choice was archeology, he went on to study medicine at the University of Basel. While working under the famous neurologist Krafft-Ebing, he settled on psychiatry as his career.

After graduating, he took a position at the Burghöltzli Mental Hospital in Zurich under Eugene Bleuler, an expert on (and the namer of) schizophrenia. In 1903, he married Emma Rauschenbach. He also taught classes at the University of Zurich, had a private practice, and invented word association at this time!

Long an admirer of Freud, he met him in Vienna in 1907. The story goes that after they met, Freud canceled all his appointments for the day, and they talked for 13 hours straight, such was the impact of the meeting of these two great minds! Freud eventually came to see Jung as the crown prince of psychoanalysis and his heir apparent.

But Jung had never been entirely sold on Freud's theory. Their relationship began to cool in 1909, during a trip to America. They were entertaining themselves by analyzing each others' dreams (more fun, apparently, than shuffleboard), when Freud seemed to show an excess of resistance to Jung's efforts at analysis. Freud finally said that they'd have to stop because he was afraid he would lose his authority! Jung felt rather insulted.

World War I was a painful period of self-examination for Jung. It was, however, also the beginning of one of the most interesting theories of personality the world has ever seen.

After the war, Jung traveled widely, visiting, for example, tribal people in Africa, America, and India. He retired in 1946, and began to retreat from public attention after his wife died in 1955. He died on June 6, 1961, in Zurich.

Ego, personal unconcious, and collective unconscious

Jung's theory divides the psyche into three parts. The first is the **ego**, which Jung identifies with the conscious mind. Closely related is the **personal unconscious**, which includes anything which is not presently conscious, but can be. The personal unconscious is like most people's understanding of the unconscious in that it includes both memories that are easily brought to mind and those that have been suppressed for some reason. But it does not include the instincts that Freud would have it include.

But then Jung adds the part of the psyche that makes his theory stand out from all others: the **collective unconscious**. You could call it your "psychic inheritance." It is the reservoir of our experiences as a species, a kind of knowledge we are all born with. And yet we can never be directly conscious of it. It influences all of our experiences and behaviors, most especially the emotional ones, but we only know about it indirectly, by looking at those influences.

There are some experiences that show the effects of the collective unconscious more clearly than others: The experiences of love at first sight, of deja vu (the feeling that you've been here before), and the immediate recognition of certain symbols and the meanings of certain myths, could all be understood as the sudden conjunction of our outer reality and the inner reality of the collective unconscious. Grander examples are the creative experiences shared by artists and musicians all over the world and in all times, or the spiritual experiences of mystics of all religions, or the parallels in dreams, fantasies, mythologies, fairy tales, and literature.

A nice example that has been greatly discussed recently is the near-death experience. It seems that many people, of many different cultural backgrounds, find that they have very similar recollections when they are brought back from a close encounter with death. They speak of leaving their bodies, seeing their bodies and the events surrounding them clearly, of being pulled through a long tunnel towards a bright light, of seeing deceased relatives or religious figures waiting for them, and of their disappointment at having to leave this happy scene to return to their bodies. Perhaps we are all "built" to experience death in this fashion.

Archetypes

The contents of the collective unconscious are called **archetypes**. Jung also called them dominants, imagos, mythological or primordial images, and a few other names, but archetypes seems to have won out over these. An archetype is an unlearned tendency to experience things in a certain way.

The archetype has no form of its own, but it acts as an "organizing principle" on the things we see or do. It works the way that instincts work in Freud's theory: At first, the baby just wants something to eat, without knowing what it wants. It has a rather indefinite yearning which, nevertheless, can be satisfied by some things and not by others. Later, with experience, the child begins to yearn for something more specific when it is hungry -- a bottle, a cookie, a broiled lobster, a slice of New York style pizza.

The archetype is like a black hole in space: You only know it's there by how it draws matter and light to itself.

The mother archetype

The **mother archetype** is a particularly good example. All of our ancestors had mothers. We have evolved in an environment that included a mother or mother-substitute. We would never have survived without our connection with a nurturing-one during our times as helpless infants. It stands to reason that we are "built" in a way that reflects that evolutionary environment: We come into this world ready to want mother, to seek her, to recognize her, to deal with her.

So the mother archetype is our built-in ability to recognize a certain relationship, that of "mothering." Jung says that this is rather abstract, and we are likely to project the archetype out into the world and onto a particular person, usually our own mothers. Even when an archetype doesn't have a particular real person available, we tend to personify the archetype, that is, turn it into a mythological "story-book" character. This character symbolizes the archetype.

(To the right is one of his many drawings from the famous Red Book, which wasn't published until 2009.)

The mother archetype is symbolized by the primordial mother or "earth mother" of mythology, by Eve and Mary in western traditions, and by less personal symbols such as the church, the nation, a forest, or the ocean. According to Jung, someone whose own mother failed to satisfy the demands of the archetype may well be one that spends his or her life seeking comfort in the church, or in identification with "the motherland," or in meditating upon the figure of Mary, or in a life at sea.

Of the more important archetypes, we have the **shadow**, which represents our animal ancestry and is often the locus of our concerns with evil and our own "dark side;" there's the **anima**, representing the female side of men, and the **animus**, representing the male side of women; and the **persona**, which is the surface self, that part of us we allow others to see.

Other archetypes include father, child, family, hero, maiden, animal, wise old man, the hermaphrodite, God, and the first man.

The self

The goal of life is to realize the **self**. The self is an archetype that represents the transcendence of all opposites, so that every aspect of your personality is expressed equally. You are then neither and both male and female, neither and both ego and shadow, neither and both good and bad, neither and both conscious and unconscious, neither and both an individual and the whole of creation. And yet, with no oppositions, there is no energy, and you cease to act. Of course, you no longer need to act.

To keep it from getting too mystical, think of it as a new center, a more balanced position, for your psyche. When you are young, you focus on the ego and worry about the trivialities of the persona. When you are older (assuming you have been developing as you should), you focus a little deeper, on the self, and become closer to all people, all life, even the universe itself. The self-realized person is actually less selfish.

The Myers-Briggs test

Katharine Briggs and her daughter Isabel Briggs Myers found Jung's ideas about people's personalities so compelling that they decided to develop a paper-and-pencil test. It came to be called the **Myers-Briggs Type Indicator**, and is one of the most popular, and most studied, tests around.

On the basis of your answers on about 125 questions, you are placed in one of sixteen types, with the understanding that some people might find themselves somewhere between two or three types. What type you are says quite a bit about you -- your likes and dislikes, your likely career choices, your compatibility with others, and so on. People tend to like it quite a bit. It has the unusual quality among personality tests of not being too judgmental: None of the types is terribly negative, nor are any overly positive. Rather than assessing how "crazy" you are, the "Myers-Briggs" simply opens up your personality for exploration.

The test has four scales. **Extroversion - Introversion** (E-I) is the most important. Test researchers have found that about 75 % of the population is extroverted.

The next one is **Sensing - Intuiting** (S-N), with about 75 % of the population sensing.

The next is **Thinking - Feeling** (T-F). Although these are distributed evenly through the population, researchers have found that two-thirds of men are thinkers, while two-thirds of women are feelers. This might seem like stereotyping, but keep in mind that feeling and thinking are both valued equally by Jungians, and that one-third of men are feelers and one-third of women are thinkers. Note, though, that society does value thinking and feeling differently, and that feeling men and thinking women often have difficulties dealing with people's stereotyped expectations.

The last is **Judging - Perceiving** (J-P), not one of Jung's original dimensions. Myers and Briggs included this one in order to help determine which of a person's functions is superior. Generally, judging people are more careful, perhaps inhibited, in their lives. Perceiving people tend to be more spontaneous, sometimes careless. If you are an extrovert and a "J," you are a thinker or feeler, whichever is stronger. Extroverted and "P" means you are a senser or intuiter. On the other hand, an introvert with a high "J" score will be a senser or intuiter, while an introvert with a high "P" score will be a thinker or feeler. J and P are equally distributed in the population.

Discussion

Quite a few people find that Jung has a great deal to say to them. They include writers, artists, musicians, film makers, theologians, clergy of all denominations, students of mythology, and, of course, some psychologists. Examples that come to mind are the mythologist Joseph Campbell, the film maker George Lucas, and the science fiction author Ursula K. Le Guin. Anyone interested in creativity, spirituality, psychic phenomena, the universal, and so on will find in Jung a kindred spirit.

But scientists, including most psychologists, have a lot of trouble with Jung. Not only does he fully support the teleological view (as do most personality theorists), but he goes a step further and talks about the mystical interconnectedness of synchronicity. Not only does he postulate an unconscious, where things are not easily available to the empirical eye, but he postulates a collective unconscious that never has been and never will be conscious.

In fact, Jung takes an approach that is essentially the reverse of the mainstream's reductionism: Jung begins with the highest levels - even spiritualism - and derives the lower levels of psychology and physiology from them.

Even psychologists who applaud his teleology and antireductionist position may not be comfortable with him. Like Freud, Jung tries to bring everything into his system. He has little room for chance, accident, or circumstances. Personality - and life in general - seems "over-explained" in Jung's theory.

I have found that his theory sometimes attracts students who have difficulty dealing with reality. When the world, especially the social world, becomes too difficult, some people retreat into fantasy. Some, for example, become couch potatoes. But others turn to complex ideologies that pretend to explain everything. Some get involved in Gnostic or Tantric religions, the kind that present intricate rosters of angels and demons and heavens and hells, and endlessly discuss symbols. Some go to Jung. There is nothing intrinsically wrong with this; but for someone who is out of touch with reality, this is hardly going to help.

These criticisms do not cut the foundation out from under Jung's theory. But they do suggest that some careful consideration is in order.

Alfred Adler

Alfred Adler was born in the suburbs of Vienna on February 7, 1870, the third child, second son, of a Jewish grain merchant and his wife. As a child, Alfred developed rickets, which kept him from walking until he was four years old. At five, he nearly died of pneumonia. It was at this age that he decided to be a physician.

Alfred was an average student and preferred playing outdoors to being cooped up in school. He was quite outgoing, popular, and active, and was known for his efforts at outdoing his older brother, Sigmund.

He received a medical degree from the University of Vienna in 1895. During his college years, he became attached to a group of socialist students, among which he found his wife-to-be, Raissa Timofeyewna Epstein. She was an intellectual and social activist who had come from Russia to study in Vienna. They married in 1897 and eventually had four children, two of whom became psychiatrists.

He began his medical career as an opthalmologist, but he soon switched to general practice, and established his office in a lower-class part of Vienna, across from the Prader, a combination amusement park and circus. His clients included circus people, and it has been suggested (Furtmuller, 1964) that the unusual strengths and weaknesses of the performers led to his insights into **organ inferiorities** and **compensation**.

He then turned to psychiatry, and in 1907 was invited to join Freud's discussion group. After writing papers on organic inferiority, which were quite compatible with Freud's views, he wrote, first, a paper concerning an **aggression instinct**, which Freud did not approve of, and then a paper on children's feelings of inferiority, which suggested that Freud's sexual notions be taken more metaphorically than literally.

Although Freud named Adler the president of the Viennese Analytic Society and the co-editor of the organization's newsletter, Adler didn't stop his criticism. A debate between Adler's supporters and Freud's was arranged, but it resulted in Adler, with nine other members of the organization, resigning to form the Society for Free Psychoanalysis in 1911. This organization became The Society for Individual Psychology in the following year.

During World War I, Adler served as a physician in the Austrian Army, first on the Russian front, and later in a children's hospital. He saw first hand the damage that war does, and his thought turned increasingly to the concept of **social interest**. He felt that if humanity was to survive, it had to change its ways!

After the war, he was involved in various projects, including clinics attached to state schools and the training of teachers. In 1926, he went to the United States to lecture, and he eventually accepted a visiting position at the Long Island College of Medicine. In 1934, he and his family left Vienna forever. On May 28, 1937, during a series of lectures at Aberdeen University, he died of a heart attack.

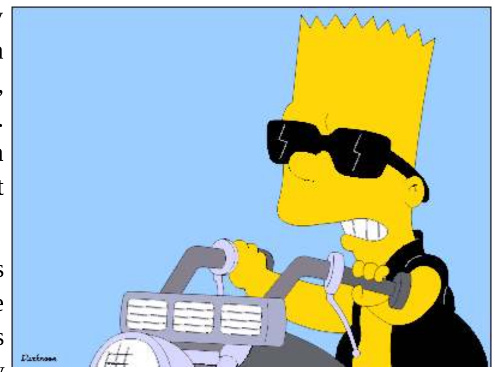
Striving

Alfred Adler postulates a single "drive" or motivating force behind all our behavior and experience. By the time his theory had gelled into its most mature form, he called that motivating force the **striving for perfection**. It is the desire we all have to fulfill our potentials, to come closer and closer to our ideal. It is, as many of you will already see, very similar to the more popular idea of self-actualization.

"Perfection" and "ideal" are troublesome words, though. On the one hand, they are very positive goals. Shouldn't we all be striving for the ideal? And yet, in psychology, they are often given a rather negative connotation. Perfection and ideals are, practically by definition, things you can't reach. Many people, in fact, live very sad and painful lives trying to be perfect! As you will see, other theorists, like Karen Horney and Carl Rogers, emphasize this problem. Adler talks about it, too. But he sees this negative kind of idealism as a perversion of the more positive understanding. We will return to this in a little while.

Striving for perfection was not the first phrase Adler used to refer to his single motivating force. His earliest phrase was the **aggression drive**, referring to the reaction we have when other drives, such as our need to eat, be sexually satisfied, get things done, or be loved, are frustrated. It might be better called the assertiveness drive, since we tend to think of aggression as physical and negative. But it was Adler's idea of the aggression drive that first caused friction between him and Freud. Freud was afraid that it would detract from the crucial position of the sex drive in psychoanalytic theory. Despite Freud's dislike for the idea, he himself introduced something very similar much later in his life: the death instinct.

Another word Adler used to refer to basic motivation was **compensation**, or striving to overcome. Since we all have problems, short-comings, inferiorities of one sort or another, Adler felt, earlier in his writing, that our personalities could be accounted for by the ways in which we do - or don't - compensate or overcome those problems. The idea still plays an important role in his theory, as you will see, but he rejected it as a label for the basic motive because it makes it sound as if it is your problems that cause you to be what you are.



One of Adler's earliest phrases was **masculine protest**. He noted something pretty obvious in his culture (and by no means absent from our own): Boys were held in higher esteem than girls. Boys wanted, often desperately, to be thought of as strong, aggressive, in control - i.e. "masculine" - and not weak, passive, or dependent - i.e. "feminine." The point, of course, was that men are somehow basically better than women. They do, after all, have the power, the education, and apparently the talent and motivation needed to do "great things," and women don't.

You can still hear this in the kinds of comments older people make about little boys and girls: If a baby boy fusses or demands to have his own way (masculine protest!), they will say he's a natural boy; If a little girl is quiet and shy, she is praised for her femininity; If, on the other hand, the boy is quiet and shy, they worry that he might grow up to be a sissy; Or if a girl is assertive and gets her way, they call her a "tomboy" and will try to reassure you that she'll grow out of it!

But Adler did not see men's assertiveness and success in the world as due to some innate superiority. He saw it as a reflection of the fact that boys are encouraged to be assertive in life, and girls are discouraged. Both boys and girls, however, begin life with the capacity for "protest!" Because so many people misunderstood him to mean that men are, innately, more assertive, he limited his use of the phrase.

The last phrase he used, before switching to striving for perfection, was **striving for superiority**. His use of this phrase reflects one of the philosophical roots of his ideas: Friederich Nietzsche developed a philosophy that considered the will to power the basic motive of human life. Although striving for superiority does refer to the desire to be better, it also contains the idea that we want to be better than others, rather than better in our own right. Adler later tended to use striving for superiority more in reference to unhealthy or neurotic striving.

Life style

A lot of this playing with words reflects Adler's groping towards a really different kind of personality theory than that represented by Freud's. Freud's theory was what we nowadays would call a reductionistic one: He tried most of his life to get the concepts down to the physiological level. Although he admitted failure in the end, life is nevertheless explained in terms of basic physiological needs. In addition, Freud tended to "carve up" the person into smaller theoretical concepts - the id, ego, and superego - as well.

Adler was influenced by the writings of Jan Smuts, the South African philosopher and statesman. Smuts felt that, in order to understand people, we have to understand them more as unified wholes than as a collection of bits and pieces, and we have to understand them in the context of their environment, both physical and social. This approach is called **holism**, and Adler took it very much to heart.

First, to reflect the idea that we should see people as wholes rather than parts, he decided to label his approach to psychology **individual psychology**. The word individual means literally "un-divided."

Second, instead of talking about a person's personality, with the traditional sense of internal traits, structures, dynamics, conflicts, and so on, he preferred to talk about **style of life** (nowadays, "lifestyle"). Life style refers to how you live your life, how you handle problems and interpersonal relations. Here's what he himself had to say about it: "The style of life of a tree is the individuality of a tree expressing itself and molding itself in an environment. We recognize a style when we see it against a background of an environment different from what we expect, for then we realize that every tree has a life pattern and is not merely a mechanical reaction to the environment."

Teleology

The last point -- that lifestyle is "not merely a mechanical reaction" -- is a second way in which Adler differs dramatically from Freud. For Freud, the things that happened in the past, such as early childhood trauma, determine what you are like in the present. Adler sees motivation as a matter of moving towards the future, rather than being driven, mechanistically, by the past. We are drawn towards our goals, our purposes, our ideals. This is called **teleology**.

Moving things from the past into the future has some dramatic effects. Since the future is not here yet, a teleological approach to motivation takes the necessity out of things. In a traditional mechanistic approach, cause leads to effect: If a, b, and c happen, then x, y, and z must, of necessity, happen. But you don't have to reach your goals or meet your ideals, and they can change along the way. Teleology acknowledges that life is hard and uncertain, but it always has room for change!

Another major influence on Adler's thinking was the philosopher Hans Vaihinger, who wrote a book called **The Philosophy of "As If."** Vaihinger believed that ultimate truth would always be beyond us, but that, for practical purposes, we need to create partial truths. His main interest was science, so he gave as examples such partial truths as protons and electrons, waves of light, gravity as distortion of space, and so on. Contrary to what many of us non-scientists tend to assume, these are not things that anyone has seen or proven to exist: They are useful constructs. They work for the moment, let us do science, and hopefully will lead to better, more useful constructs. We use them "as if" they were true. He called these partial truths **fictions**.

Vaihinger, and Adler, pointed out that we use these fictions in day to day living as well. We behave as if we knew the world would be here tomorrow, as if we were sure what good and bad are all about, as if everything we see is as we see it, and so on. Adler called this **fictional finalism**. You can understand the phrase most easily if you think about an example: Many people behave as if there were a heaven or a hell in their personal future. Of course, there may be a heaven or a hell, but most of us don't think of this as a proven fact. That makes it a "fiction" in Vaihinger's and Adler's sense of the word. And finalism refers to the teleology of it: The fiction lies in the future, and yet influences our behavior today.

Adler added that, at the center of each of our lifestyles, there sits one of these fictions, an important one about who we are and where we are going.

Discussion

Criticisms of Adler tend to involve the issue of whether or not, or to what degree, his theory is scientific. The mainstream of psychology today is experimentally oriented, which means, among other things, that the concepts a theory uses must be measurable and manipulable. This in turn means that an experimental orientation prefers physical or behavioral variables. Adler, as you saw, uses basic concepts that are far from physical and behavioral: Striving for perfection? How do you measure that? Or compensation? Or feelings of inferiority? Or social interest? The experimental method also makes a basic assumption: That all things operate in terms of cause and effect. Adler would certainly agree that physical things do so, but he would adamantly deny that people do! Instead, he takes the teleological route, that people are "determined" by their ideals, goals, values, "final fictions." Teleology takes the necessity out of things: A person doesn't have to respond a certain way to a certain circumstance; A person has choices to make; A person creates his or her own personality or lifestyle. From the experimental perspective, these things are illusions that a scientist, even a personality theorist, dare not give in to.

CHAPTER OVERVIEW

4: Making Psychology a Science

4.1: Wundt

Read section 3 and section 7!

Wundt

4.2: Early Psychological Laboratories

James McKeen Cattell (1928)

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Laboratories for research and teaching in the sciences are of comparatively recent origin. They may be regarded as part of the industrial revolution, for there is a close parallel in causes and effects between the development of the factory system and of scientific laboratories. The industrial revolution began with the exploitation by machinery of coal and iron in England; it may perhaps be dated from the use of the steam engine of Watts in the coal mines of Cornwall about a hundred and fifty years ago.

The laboratory had its origin fifty years later in Germany as part of the scientific renaissance following the Napoleonic wars. The University of Berlin was founded by Wilhelm von Humboldt and Frederich William III in 1810. The first laboratory of chemistry was opened by Justus von Liebig at Giessen in 1824. This was followed by similar laboratories at Göttingen under Wöhler in 1836, at Marburg under Bunsen in 1840, and at Leipzig under Erdmann in 1843. The first English laboratory was the College of Chemistry, now part of the Imperial College of Science and Technology of the University of London, which was opened in 1845 by von Hoffmann, brought from Germany by Prince Albert. Benjamin Silliman founded at Yale University the first American laboratory for the teaching of chemistry.

Prior to the industrial revolution the artisan worked at home, sometimes with 'prentices, who were often his children. The factories, the mines and the systems of transportation, with their machinery, their skilled overseers and division of labor, their owners and entrepreneurs, their exchange of commodities and ideas, created a remarkable economy in production, so that now each individual may perhaps work half as long and consume twice as much wealth as formerly.

But there are serious drawbacks in the lack of freedom and initiative of the workman, in the loss of joy in creative work. The situation in the laboratory is similar. A professor may have many associates, assistants and students; expensive apparatus and extensive libraries may be installed; division of labor in each laboratory and among laboratories can be planned; there may be exchange of ideas and of information on the progress of research; students are taught in large groups. Production is greatly increased, perhaps quadrupled, as in the industrial system, But the scientific man is subject to administrative controls; he is no longer free; he must compromise with others and teach all sorts of students. The system is useful for the production of a large mass of routine work; it may not be favorable to creative genius.

Anatomy has been called the mother of the sciences; dissecting rooms go back to the medieval universities of Italy. Observatories, museums, botanical gardens, academies of science and university schools, where research was undertaken and in which students and assistants were taught and trained, preceded organized laboratories. Chemistry is the gold transmuted through alchemy; we have all seen on the stage the laboratory of Faust. Christian fathers say that when "the sons of God saw the daughters of men, that they were fair and they took them wives," as told in the sixth chapter of Genesis, these fallen angels taught the fair daughters of men the arts of astrology and alchemy. Scientific men who do not care for special creations may assume that there has been a gradual development from the time of the first experiment by an anthropoid ape, or it may be by a paramecium or an electron. If, however, we want an official beginning for the first scientific laboratory, it will be the laboratory of chemistry at Giessen, the hundredth anniversary of whose foundation was celebrated three years ago.

Chemical laboratories were followed by laboratories of physics and biology. I worked in the first American biological laboratory in its early days. It was established at the Johns Hopkins University by Newell Martin, a student of Huxley, who at the Royal College of Science had founded the first laboratory of biology. From the laboratories of Martin and Brooks at the Johns Hopkins have proceeded many of our most eminent biological workers. The Johns Hopkins also led in the establishment under Welch, Mall, Abel and Howell of laboratories in the medical sciences. But there is obviously no sharp line of demarcation between the modern laboratory and earlier groups of workers, such as the great school of zoology conducted by Agassiz at Harvard.

The first laboratory of psychology was established by Wilhelm Wundt. In an article on the Leipzig laboratory, published in *Mind* in 1888 and submitted to Professor Wundt, I give the date as 1879. The fiftieth anniversary of the founding of the laboratory [p. 544] was, however, celebrated at Leipzig in 1926. Wundt published his "Grundzüge der physiologischen Psychologie" in 1874 and was called from Zürich to a chair of philosophy at Leipzig in 1875. The *Psychologische Institut* there was a gradual development. Wundt writes in his autobiography "Erlebtes und Erkanntes," published in 1920, that Kraepelin, Lehmann and I were his three earliest "Arbeitsgenossen" who remained faithful to psychology and that we worked with him at a time when the institute was his private undertaking and lacked official recognition on the part of the university. The first research published from the Leipzig laboratory was apparently a doctor's dissertation by Dr. Max Friedrich carried out during the winter semester of 1879-80.

Wundt writes in the preface to the "Physiologische Psychologie" that it undertakes "ein neues Gebiet der Wissenschaft abzugrenzen," but he was partly anticipated by Hermann Lotze, whose "Medizinische Psychologie" was published in 1852. Both Lotze and Wundt had a medical education and were professors of philosophy. Their books are landmarks in the history of our science. It was my privilege to hear the last course of lectures on psychology by Lotze given at Göttingen in the winter semester of 1880-81. In accordance with the custom of that university Lotze dictated summaries which could be written down verbatim even by one who had small psychology and less German. The "Dictata" of that year were published and have been translated into English. In the spring of 1881 Lotze, then 74 years of age, migrated to Berlin and died, according to Göttingen opinion, of homesickness.

Herbart, whom Lotze succeeded at Göttingen, had tried to give a mathematical formulation to psychology as Spinoza had to philosophy. He published the first edition of his "Einleitung in die Philosophie" in 1813. There followed Drobitsch, Lindner, Benecke, Volkmar and other German psychologists. In England we have the notable development of association and analytic psychology from Locke through Berkeley, Hume, the Mills and Bain to Ward. The first edition of Carpenter's "Mental Physiology," to-day a useful and readable book, was published in 1874, the same year as Wundt's "Grundzüge." In England and in France there were numerous workers in the fields of physiological and pathological psychology.

The most important developments for laboratory psychology were through the great German physiologists and physicists, most of all Helmholtz, who passed from physiology to physics. His "Physiologische Optik," recently translated under the editorship of Professor Southall and published as an act of piety by the Optical Society, and his "Tonempfindingen," of which there is an earlier translation, are classics in the history of science. E. H. Weber became professor of anatomy at Leipzig in 1818 at the age of twenty-three, being later transferred to physiology. The law that bears his name was stated in his "Annotationes," published from 1834 to 1851. Fechner was appointed professor of physics at the same university in 1834; his "Zendavesta" was published in 1851, his "Elemente der Psychophysik" in 1860. When I was a student at Leipzig he was over eighty-five years old and blind from experiments on vision, a charming man, intensely interested in his psychophysical experiments, though chiefly in philosophical interpretations.

The middle fifty years of the last century were the golden age of the German university and of science, its *Wunderkind*. It is marvelous what was accomplished then and there. Thus in the little corner of the field of science concerned with the psychology of the sense of vision there worked, in addition to Helmholtz and Fechner, a notable company, including Aubert, Brücke, du Bois-Reymond, Donders (in Holland), Exner, Fraunhofer, Fick, von Graefe (who examined my eyes when I was a child of eight), Hering, Hermann, von Kries, Listing, Johann Mueller, Nagel, Purkinje, Vierordt, the Webers and many more. There is no such group in the world to-day working on vision or in any other part of experimental psychology. At that time the investigation of the other senses, of movement, of the time of reaction and much else was pursued probably to greater effect than in all the innumerable laboratories of to-day.

The fields so fertile in the nineteenth century were of course cleared at an earlier time. Experiments on vision go back to Kepler, Huygens and Newton. Weber's law was anticipated by Bouguer and Lambert; Fechner's law by Bernouilli[*sic*] and Laplace; the personal equation by the astronomers. Observations on after-images were made not only by Goethe, the elder Darwin, Buffon and Newton among others, but also by Augustine and Aristotle. Very curiously the problems of psychological measurement were clearly stated by the poet Shelley, who more than a hundred years ago wrote: "A scale might be formed, graduated according to the degrees of a combined scale of intensity, duration, connexion, periods of recurrence, and utility, which would be the standard, according to which all ideas might be measured." When I came across this passage in Shelley it seemed almost incredible that he of all men should have written it, as indeed it is that the most unearthly of poets should have been the son of a country squire. But England has always given birth to great men in families and as sports. It has been said that Graham Bell -- he too was British -- could not have invented the telephone if he had been a physicist, for he would have known that it was impossible; so it may be said [p. 545] that Francis Galton could not have accomplished his great work toward founding modern psychology if he had been a psychologist, for he would have known that it was not psychology. Galton, like Darwin, his cousin, had no university position and no laboratory. He published his "Hereditary Genius" in 1869, his "Inquiries into Human Faculty" in 1883.

With intermissions I was a student at Leipzig under Wundt from 1881 to 1886, serving during the last year as laboratory assistant in psychology, the first to be appointed there or anywhere. Wundt had a higher opinion, doubtless with good reason, of American enterprise than of American scholarship. In his reminiscences he writes that with "bekannter amerikanischer Entschlossenheit" I approached him and declared: "Herr Professor, you need an assistant and I shall be your assistant" He was the most kindly of men and was much worried lest I should not pass my doctorate examination in physics under Hankel and in zoology under Leuckhart, but these distinguished professors also fortunately made due allowance for a child of the wilderness. Wundt's combined courtesy and remoteness from the modern world may be illustrated by an incident. At that time women were seldom admitted to university lectures, but at my request he gave permission to an American of fine intelligence to attend his course on psychology, which was frequented by two or three hundred students, among them the most stupid in the university, for all theological students were required to attend. One day he said: "I am sorry that I let Miss X attend my lectures; it embarrasses me; I feel that I ought to speak in a way that a woman can understand."

When I showed Wundt an outline of the work that I proposed for a doctor's thesis on the reaction-time, including complicated responses and a study of individual differences his comment was: "ganz Amerikanisch." As a matter of fact I did the work in my own rooms and with my own apparatus. At that time students were expected to work in the laboratory on a subject assigned by the professor, during certain definite hours in the afternoon and with the apparatus supplied, which had to be put away neatly in the cases after a two-hour period. We used two batteries of Daniel cells and when these were set up and got into running order it was nearly time to take them apart, wash the zincs and coppers and put the fluids into bottles. As in this process we were likely to splash sulphuric acid on our clothes we kept handy a bottle of ammonia, which was very promptly applied to the stains. At that time I anticipated Dr. Watson in an observation on the conditioned reflex, for when the German student who worked with me drew a mouthful of dilute sulphuric acid through the syphon that we used, he immediately reached for the ammonia bottle and took a mouthful of that.

In the early eighties Wundt's laboratory was housed on the top floor of the *Convict* building, where indigent students had their meals. He used to walk through the laboratory after his lecture, always courteous and ready to answer questions, but, as I remember it, usually limiting his visit to five or ten minutes. He was interested in the laboratory as a system, and as a method of introspection, but he was not himself a laboratory worker. His interests were very broad. His "Logik," published from 1880 to 1883, contains in the second edition 1,995 pages; his "Ethik," also published while I was at Leipzig, contains in its third edition 933 pages. The "System der Philosophie" published in 1889, contains in its third edition 738 pages. The last edition of the "Physiologische Psychologie" contains 2,317 pages, the "Völkerpsychologie," 3,161 pages. And they are very large pages.

These books and others Wundt composed on a typewriter that I gave him, one of the first in Germany. Avenarius once remarked that I had by this gift done a serious disservice to philosophy, for it had enabled Wundt to write twice as many books as would otherwise have been possible. At that time the relations of German professors were curious from an American point of view, Wundt was not in friendly relations with Helmhoitz, Stumpf, Müller and others. Stumpf, next to Wundt the most distinguished of German psychologists, was professor at Halle, only three quarters of an hour by train from Leipzig, and Wundt was asked for an introduction. He said that he was sorry that he could not give it, as he did not know Professor Stumpf personally; it was better so, for they could then write more freely when there was a difference of opinion -- and they did a couple of years later.

At the beginning of the semester students who wanted to undertake experimental work stood before Wundt in a row and from a slip of paper that he held in his hand he assigned topics in order. The year that I appeared there were six or seven of us, representing nearly as many nationalities. I was given the problem of reacting to colored lights; first when the light was seen, and second when the color was distinguished, and by subtracting one time from the other of obtaining what Wundt called the "Apperceptionszeit." This I could not do, but the problem was most useful to me, for it led me to realize the limitations of introspection and to base my work on objective measurements of behavior. Wundt's refusal to admit any subject to the laboratory except a psychologist who could use the results introspectively was [p. 546] also useful, for it led me to transfer the work to my rooms and make there the first psychological measurements of individual differences and to attempt to develop the useful applications of psychology -- with both of which efforts Wundt had no sympathy.

Wundt rejected as a doctorate dissertation Münsterberg's very able monograph on "Die Willenshandlungen" because it did not coincide with his own theories. He calls Stanley Hall's excellent sketch of his life and work an "erdichtete Biographie die von anfang bis zu Ende erfunden ist." But such things were only the righteous indignation of the Hebrew prophet denouncing the enemies of the Lord. The academic life in Germany in those days was exalted. The nation, the university, the professor, were sacrosanct. It was a fine experience to be admitted to the outer court of the temple before the money changers had entered. Wundt himself was the ideal German professor, with boundless learning shading toward the pedantic, fully conscious of his plenary inspiration, yet withal most modest, shy and kindly; a seer before his students, a child at home, a truly great man.

Wundt's laboratory of psychology was international in its reputation and influence, attracting students from all parts of the world, Americans and Russians predominating. In 1892 it received larger quarters and in 1897 was removed to one of the buildings vacated by the Medical School, where fourteen rooms were remodeled for its purposes. In the late eighties there were beginnings of laboratories under Ebbinghaus at Berlin, under Müller at Göttingen, and under students of Wundt who were my contemporaries and friends, Münsterberg st Freiburg, Martius at Bonn, and Lehmann at Copenhagen.

The second laboratory of psychology was organized by G. Stanley Hall at the Johns Hopkins University early in the year 1883. I was there before Hall, holding a fellowship in philosophy, this award for a thesis on Lotze having been made by the professor of Latin, who knew even less about philosophy than I did, or the fellowship would have been given to John Dewey. He was there as a student, as were also Joseph Jastrow and H. H. Donaldson. We helped Hall set up a modest laboratory in a private house adjacent to the center of ugly little brick buildings and great men that formed the university. The small group of professors working there included Remsen, Rowland, Sylvester, Gildersleeve, Haupt, Adams, Brooks and Martin.

It is a curious fact that neither of the founders of our first two psychological laboratories was a laboratory worker. Hall's chair, like Wundt's, was not limited to psychology; he lectured on philosophy and he also conducted courses in pedagogy. The range of his interests was large, but it was the human aspects of life that he cared for rather than abstract quantitative measurements. Like James he was a man of literary genius swayed by the emotions, which are such a large part of life and as yet such a small part of our science. Minot, the distinguished Harvard embryologist, once said that he envied my occupation with a science concerned with human interests. My reply was that my experiments had as little to do with such things as his had with love and children. Hall wrote about children, adolescence and senescence, religion and sex, the drama of life. He and James were giants in the land, overtowering their descendants of a work-a-day world. As Wundt established the *Philosophische Studien* to publish the work from his laboratory and his own articles on psychology and philosophy, so Hall established the *American Journal of Psychology*. The early volumes give a survey of the work done in Baltimore, which was largely physiological and psychiatrical. Hall was much interested in insanity and other pathological aspects of psychology and we used to go regularly to the Bayview Hospital for the Insane. These interests were maintained and in the last conversation that I had with him in his lonely house at Worcester he wanted especially to know why orthodox *American* psychologists cared so little for Freud and psychoanalysis. He showed me a mass of publications and notes that he had collected on the subject.

Hall was called upon to organize Clark University in 1888 and gathered there a group of outstanding scientific men, including Michelson, Webster, Bolza, Neff, Whitman, Mall, Donaldson, Lombard, McMurrich and Boas. The financial support of the university by Mr. Clark was less liberal than had been anticipated and Dr. Harper took over in a body a large part of these men for the faculty of the new University of Chicago. In his "Life and Confessions" Hall remarks: "I felt his act comparable to that of a housekeeper who would steal in at the back door to engage servants at a higher price." Sanford went with Hall from the Johns Hopkins to Clark and became director of the laboratory of psychology which was opened in 1889. The Johns Hopkins laboratory was closed and the apparatus dispersed until it was reestablished by Professor Baldwin and Professor Stratton. Hall and Clark University long maintained a dominant position in psychology and the psychological side of education. In his death there ends the romantic and heroic era of our science.

The laboratory of psychology at the University of Pennsylvania was founded by me in 1887, though it was only in January, 1889, that a special laboratory with adequate equipment of apparatus was opened. The laboratories at Leipzig and the Johns Hopkins were for research, and psychology was only part of the field covered by the professor. At the University [p. 547] of Pennsylvania a professorship of psychology was for the first time established and laboratory courses for students were for the first time given. It might consequently be argued by a partial advocate that this was the first laboratory of psychology in the sense that Liebig's chemical laboratory at Giessen was the first scientific laboratory. More significant is the circumstance that in this laboratory the research work and the courses for students were based on objective measurements of responses to the environment with special reference to individual and group differences and to the useful applications of psychology, thus leading to the development of modern educational, clinical and industrial psychology.

In 1888 I was also lecturer at Bryn Mawr College and at the University of Cambridge, conducting in both institutions laboratory courses for students. At Cambridge the work was in conjunction with the lectures of Professor James Ward and in the Cavendish laboratory of which the present Sir Joseph Thomson was the director, having just before, at the age of twenty-six, succeeded Maxwell and Rayleigh in the professorship of physics. In the Cavendish laboratory was also set up apparatus for research and this was the beginning of: the first British laboratory of psychology. At that time I had the privilege of assisting Galton in the arrangement of the Anthropometric Laboratory in the South Kensington Museum and we began in cooperation the preparation of a book of instructions for a laboratory course in psychology.

The five-year period from 1887 to 1892 is distinguished for the development of laboratories of psychology in the United States. For earlier work tribute should in passing be paid to James McCosh, Presbyterian clergyman from Scotland and president of a Presbyterian college, who at Princeton promoted the study of organic evolution and physiological psychology. George Trumbull

Ladd, also a clergyman, was called to Yale as professor of philosophy in 1881 and developed there courses in physiological psychology, leading to the publication in 1887 of his "Outlines of Physiological Psychology." With James and Hall he shares the honor of leading in the development of psychology in America. The laboratory at Yale was organized by Professor Ladd in 1892 with Dr. E. M. Scripture as instructor.

Work in experimental psychology leading to the establishment of a laboratory was began by Professor Joseph Jastrow at Wisconsin in 1888. His service as professor of psychology is the longest in the history of our science. A year or two later laboratories were established at Indiana University by President W. L. Bryan, at the University of Nebraska by Professor H. K. Wolfe, at Brown University by Professor E. B. Delabarre and at Stanford University by Professor Frank Angell. Professor J. Mark Baldwin was called to Toronto in 1890 and established there a psychological laboratory, as he did at Princeton when he returned to that university in 1893. In 1895 we together founded the *Psychological Review*, which, with its children, *The Psychological Monographs*, *The Psychological Index*, *The Psychological Bulletin*, the *Journal* of *Experimental Psychology*, and the newly established *Psychological Abstracts*, have now, through the generous cooperation of Professor Warren, been acquired and are being conducted by the American Psychological Association.

The professorship of psychology and the laboratory of psychology at Columbia University date from 1891. There worked Professor Thorndike, Professor Woodworth and many others who have led in the development of modern psychology. The following year is notable for the establishment of the psychological laboratories at Harvard and Cornell and the calling to America of Hugo Münsterberg and E. B. Titchener. At Cornell the traditions of the Leipzig laboratory have been best maintained. Titchener brought to us the scholarship of the Oxford don and the research ideals of the German professor. Now he has followed James, Hall and Münsterberg, leaving the world more drab and empty.

Where were James, Royce and Münsterberg was the center of psychology. James was appointed professor of psychology at Harvard in 1889, having been from 1872 to 1880 instructor and assistant professor of comparative anatomy and physiology, after 1880 assistant professor of philosophy, becoming again professor of philosophy in 1897. His great work, "The Principles of Psychology," was published in 1890. In a letter addressed to me as editor of SCIENCE in 1895 James thus tells of the development of work in experimental psychology at Harvard: "I myself, 'founded' the instruction in experimental psychology at Harvard in 1874-5, or 1876, I forget which. For a long series of years the laboratory was in two rooms of the Scientific School building, which at last became choked with apparatus, so that a change was necessary. I then, in 1890, resolved on an altogether new departure, raised several thousand dollars, fitted up Dane Hall, and introduced laboratory exercises as a regular part of the undergraduate psychology-course. Dr. Herbert Nichols, then at Clark, was appointed in 1891 assistant in this part of the work; and Professor Münsterberg was made director of the laboratory in 1892."

With the publication of James's "Principles of Psychology" in 1890, the opening of the laboratories at Harvard, Yale and Cornell in 1892, and the establishment of the American Psychological Association [p. 548] in the same year, the earlier period of psychology in America may be closed. The few survivors may look back upon it as the golden age of our science, but that is doubtless due only to the presbyopia that obscures the vision of objects near at hand. In the thirty-five years that have since passed the number of our workers in psychology has increased to an extent perhaps without parallel in any other country or in any other science. We welcome the opening at Wittenberg College of a new laboratory which, under the direction of Professor Reymert, will become a new center for psychological teaching and research.

Footnotes

[1] Address on the occasion of the inauguration of the Psychological Laboratory of Wittenberg College, Springfield, Ohio, October 21, 1927.

CHAPTER OVERVIEW

5: Behaviorism and Cognitive Psychology

5.1: Behaviorism

Behaviorism is the philosophical position that says that psychology, to be a science, must focus its attentions on what is observable -- the environment and behavior -- rather than what is only available to the individual -- perceptions, thoughts, images, feelings.... The latter are subjective and immune to measurement, and therefore can never lead to an objective science.

The first behaviorists were Russian. The very first was **Ivan M. Sechenov** (1829 to 1905). He was a physiologist who had studied at the University of Berlin with famous people like Müller, Du Bois-Reymond, and Helmholtz. Devoted to a rigorous blend of associationism and materialism, he concluded that all behavior is caused by stimulation.

In 1863, he wrote **Reflexes of the Brain**. In this landmark book, he introduced the idea that there are not only **excitatory** processes in the central nervous system, but **inhibitory** ones as well.

Vladimir M. Bekhterev (1857 to 1927) is another early Russian behaviorist. He graduated in 1878 from the Military Medical Academy in St. Petersburg, one year before Pavlov arrived there. He received his MD in 1881 at the tender age of 24, then went to study with the likes of Du Bois-Reymond and Wundt in Berlin, and Charcot in France.

He established the first psychology lab in Russia at the university of Kazan in 1885, then returned to the Military Medical Academy in 1893. In 1904, he published a paper entitled "Objective Psychology," which he later expanded into three volumes.

He called his field **reflexology**, and defined it as the objective study of stimulus-response connections. Only the environment and behavior were to be discussed! And he discovered what he called the **association reflex** -- what Pavlov would call the conditioned reflex.

Ivan Pavlov

Which brings us to the most famous of the Russian researchers, Ivan Petrovich Pavlov (1849-1936). After studying for the priesthood, as had his father, he switched to medicine in 1870 at the Military Medical Academy in St. Petersburg. It should be noted that he walked from his home in Ryazan near Moscow hundreds of miles to St. Petersburg!

In 1879, he received his degree in natural science, and in 1883, his MD. He then went to study at the university of Leipzig in Germany. In 1890, he was offered a position as professor of physiology at his alma mater, the Military Medical Academy, which is where he spent the rest of his life. It was in 1900 that he began studying reflexes, especially the salivary response.

In 1904, he was awarded the Nobel Prize in physiology for his work on digestion, and in 1921, he received the Hero of the Revolution Award from Lenin himself.

Pavlovian (or **classical**) conditioning builds on reflexes: We begin with an **unconditioned stimulus** and an **unconditioned response** -- a reflex! We then associate a **neutral stimulus** with the reflex by presenting it with the unconditioned stimulus. Over a number of repetitions, the neutral stimulus by itself will elicit the response! At this point, the neutral stimulus is renamed the **conditioned stimulus**, and the response is called the **conditioned response**.

Or, to put it in the form that Pavlov observed in his dogs, some meat powder on the tongue makes a dog salivate. Ring a bell at the same time, and after a few repetitions, the dog will salivate upon hearing the bell alone -- without being given the meat powder!

Pavlov agreed with Sechenov that there was inhibition as well as excitation. When the bell is rung many times with no meat forthcoming, the dog eventually stops salivating at the sound of the bell. That's extinction. But, just give him a little meat powder once, and it is as if he had never had the behavior extinguished: He is right back to salivating to the bell. This **spontaneous recovery** strongly suggests that the habit has been there all alone. The dog had simply learned to inhibit his response.

Pavlov, of course, could therefore condition not only excitation but inhibition. You can teach a dog that he is NOT getting meat just as easily as you can teach him that he IS getting meat. For example, one bell could mean dinner, and another could mean dinner is over. If the bells, however, were too similar, or were rung simultaneously, many dogs would have something akin to a nervous breakdown, which Pavlov called an **experimental neurosis**.

In fact, Pavlov classified his dogs into four different personalities, *à la* the ancient Greeks: Dogs that got angry were choleric, ones that fell asleep were phlegmatic, ones that whined were melancholy, and the few that kept their spirits up were sanguine! The relative strengths of the dogs' abilities to activate their nervous system and calm it back down (excitation and inhibition) were the explanations. These explanations would be used later by Hans Eysenck to understand the differences between introverts and extraverts!

Another set of terms that comes from Pavlov are the first and second signal systems. The **first signal system** is where the conditioned stimulus (a bell) acts as a “signal” that an important event is to occur -- i.e. the unconditioned stimulus (the meat). The **second signal system** is when arbitrary symbols come to stand for stimuli, as they do in human language.

Edward Lee Thorndike

Over in America, things were happening as well. Edward Lee Thorndike, although technically a functionalist, was setting the stage for an American version of Russian behaviorism. Thorndike (1874-1949) got his bachelors degree from Wesleyan University in Connecticut in 1895 and his masters from Harvard in 1897. While there he took a class from William James and they became fast friends. He received a fellowship at Columbia, and got his PhD in 1898. He stayed to teach at Columbia until he retired in 1940.

He will always be remembered for his cats and his poorly constructed “puzzleboxes.” These boxes had escape mechanisms of various complexities that required that the cats do several behaviors in sequence. From this research, he concluded that there were two laws of learning:

1. **The law of exercise**, which is basically the same as Aristotle’s law of frequency. The more often an association (or neural connection) is used, the stronger the connection. Naturally, the less it is used, the weaker the connection. These two were referred to as the law of use and disuse respectively.

2. **The law of effect**. When an association is followed by a “satisfying state of affairs,” the connection is strengthened. And, likewise, when an association is followed by an unsatisfying state of affairs, it is weakened. Except for the mentalistic language (“satisfying” is not behavioral!), it is the same thing as Skinner’s operant conditioning.

In 1929, his research led him to abandon all of the above except what we would now call reinforcement (the first half of law 2).

He is also known for his study of **transfer of training**. It was believed back then (and is still often believed) that studying difficult subjects -- even if you would never use them -- was good for you because it “strengthened” your mind, sort of like exercise strengthens your muscles. It was used back then to justify making kids learn Latin, just like it is used today to justify making kids learn calculus. He found, however, that it was only the similarity of the second subject to the first that leads to improved learning in the second subject. So Latin may help you learn Italian, or algebra may help you learn calculus, but Latin won’t help you learn calculus, or the other way around.

John Broadus Watson

John Watson was born January 9, 1878 in a small town outside Greenville, South Carolina. He was brought up on a farm by a fundamentalist mother and a carousing father. When John was 12, they moved into the town of Greenville, but a year later his father left the family. John became a troublemaker and barely passed in school.

At 16, he began attending Furman University, also in Greenville, and he graduated at 22 with a Masters degree. He then went on to the University of Chicago to study under John Dewey. He found Dewey “incomprehensible” and switched his interests from philosophy to psychology and neurophysiology. Dirt poor, he worked his way through graduate school by waiting tables, sweeping the psych lab, and feeding the rats.

In 1902 he suffered from a “nervous breakdown” which had been a long time coming. He had suffered from an intense fear of the dark since childhood -- due to stories he had heard in childhood about the devil doing his work in the night -- and this exacerbated into depression.

Nevertheless, after some rest, he finished his PhD the following year, got an assistantship with his professor, the respected functionalist James Angell, and married a student in his intro psych class, Mary Ickes. They would go on to have two children. (The actress Mariette Hartley is his granddaughter.)

The following year, he was made an instructor. He developed a well-run animal lab where he worked with rats, monkeys, and terns. Johns Hopkins offered him a full professorship and a laboratory in 1908, when the previous professor was caught in a brothel.

In 1913, he wrote an article called "Psychology as a Behaviorist Views It" for **Psychological Review**. Here, he outlined the behaviorist program. This was followed in the following year by the book **Behaviorism: An introduction to comparative**

psychology. In this book, he pushed the study of rats as a useful model for human behavior. Until then, rat research was not thought of as significant for understanding human beings. And, by 1915, he had absorbed Pavlov and Bekhterev's work on conditioned reflexes, and incorporated that into his behaviorist package.

In 1917, he was drafted into the army, where he served until 1919. In that year, he came out with the book **Psychology from the Standpoint of a Behaviorist** -- basically an expansion of his original article.

At this time, he expanded his lab work to include human infants. His best known experiment was conducted in 1920 with the help of his lab assistant Rosalie Rayner. "**Little Albert**," a 9 month old child, was conditioned to fear a white rat by pairing it seven times with a loud noise made by hitting a steel bar with a hammer. His fear quickly generalized to a rabbit, a fur coat, a Santa Claus mask, and even cotton. Albert was never "deconditioned" because his mother and he moved away. It was clear, however, that the conditioning tended to disappear (extinguish) rather quickly, so we assume that Albert was soon over his fear. This suggests that conditioned fear is not really the same as a phobia. Later, another child, three year old Peter, was conditioned and then "de-conditioned" by pairing his fear of a rabbit with milk and cookies and other positive things gradually.

In that year, his affair with his lab assistant was revealed and his wife sued for divorce. The administration at Johns Hopkins asked him for his resignation. He immediately married Rosalie Rayner and began looking for business opportunities.

He soon found himself working for the V. Walter Thompson advertising agency. He worked in a great variety of positions within the company, and was made vice president in 1924. By all standards of the time, he was very successful and quite rich! He increased sales of such items as Pond's cold cream, Maxwell House coffee, and Johnson's baby powder, and is thought to have invented the slogan "LSMFT -- Lucky Strikes Means Fine Tobacco."

He published his book **Behaviorism**, designed for the average reader, in 1925, and revised it in 1930. This was his final statement of his position.

Psychology according to Watson is essentially the science of stimuli and responses. We begin with reflexes and, by means of conditioning, acquire learned responses. Brain processes are unimportant (he called the brain a "mystery box"). Emotions are bodily responses to stimuli. Thought is subvocal speech. Consciousness is nothing at all.

Most importantly, he denied the existence of any human instincts, inherited capacities or talents, and temperaments. This **radical environmentalism** is reflected in what is perhaps his best known quote:

> *Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select -- doctor, lawyer, artist, merchant-chief and, yes, even beggar-man and thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors. (In **Behaviorism**, 1930)*

In addition to writing popular articles for McCall's, Harper's, Collier's and other magazines, he published **Psychological Care of Infant and Child** in 1928. Among other things, he saw parents as more likely than not to ruin their child's healthy development, and argued particularly against too much hugging and other demonstrations of affection!

In 1936, he was hired as vice-president of another agency, William Esty and Company. He devoted himself to business until he retired ten years later. He died in New York City on September 25, 1958.

E. C. Tolman

A very different theory would also have some popularity before the behaviorism left the experimental scene to the cognitivists: The cognitive behaviorism of Edward Chase Tolman. E. C. was born April 14, 1886 in Newton, Mass. His father was a businessman, his mother a housewife and fervent Quaker. He and his older brother attended MIT. His brother went on to become a famous physicist.

E. C. was strongly influenced by reading William James, so in 1911 he went to graduate school at Harvard. While there, he spent a summer in Germany studying with Kurt Koffka, the Gestalt psychologist. He received his PhD in 1915.

He went off to teach at Northwestern University. But he was a shy teacher, and an avowed pacifist during World War I, and the University dismissed him in 1918. He went to teach at the University of California at Berkeley. He also served in the OSS (Office of Strategic Services) for two years during World War II.

The University of California required loyalty oaths of the professors there (inspired by Joseph McCarthy and the "red scare"). Tolman led protests and was summarily suspended. The courts found in his favor and he was reinstated. In 1959 he retired, and received an honorary doctorate from the same University of California at Berkeley! Unfortunately, he died the same year, on November 19.

Although he appreciated the behaviorist agenda for making psychology into a true objective science, he felt Watson and others had gone too far.

1. Watson's behaviorism was the study of "twitches" -- stimulus-response is too **molecular** a level. We should study whole, meaningful behaviors: the **molar level**.

2. Watson saw only simple cause and effect in his animals. Tolman saw **purposeful**, goal-directed behavior.

3. Watson saw his animals as "dumb" mechanisms. Tolman saw them as forming and testing **hypotheses** based on prior experience.

4. Watson had no use for internal, "mentalistic" processes. Tolman demonstrated that his rats were capable of a variety of **cognitive processes**.

An animal, in the process of exploring its environment, develops a **cognitive map** of the environment. The process is called **latent learning**, which is learning in the absence of rewards or punishments. The animals develops **expectancies** (hypotheses) which are **confirmed** or not by further experience. Rewards (and punishments) come into play only as motivators for **performance** of a learned behavior, not as the causes of learning itself.

He himself acknowledged that his behaviorism was more like Gestalt psychology than like Watson's brand of behaviorism. From our perspective today, he can be considered one of the precursors of the cognitive movement.

B. F. Skinner

Burrhus Frederic Skinner was born March 20, 1904, in the small Pennsylvania town of Susquehanna. His father was a lawyer, and his mother a strong and intelligent housewife. His upbringing was old-fashioned and hard-working.

Burrhus was an active, out-going boy who loved the outdoors and building things, and actually enjoyed school. His life was not without its tragedies, however. In particular, his brother died at the age of 16 of a cerebral aneurysm.

Burrhus received his BA in English from Hamilton College in upstate New York. He didn't fit in very well, not enjoying the fraternity parties or the football games. He wrote for the school paper, including articles critical of the school, the faculty, and even Phi Beta Kappa! To top it off, he was an atheist -- in a school that required daily chapel attendance.

He wanted to be a writer and did try, sending off poetry and short stories. When he graduated, he built a study in his parents' attic to concentrate, but it just wasn't working for him.

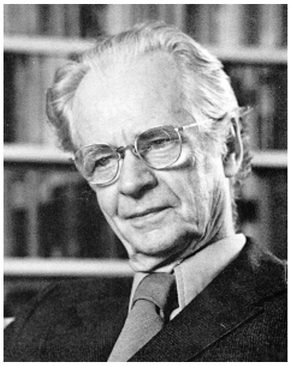
Ultimately, he resigned himself to writing newspaper articles on labor problems, and lived for a while in Greenwich Village in New York City as a "bohemian." After some traveling, he decided to go back to school, this time at Harvard. He got his masters in psychology in 1930 and his doctorate in 1931, and stayed there to do research until 1936.

Also in that year, he moved to Minneapolis to teach at the University of Minnesota. There he met and soon married Yvonne Blue. They had two daughters, the second of whom became famous as the first infant to be raised in one of Skinner's inventions, the air crib. Although it was nothing more than a combination crib and playpen with glass sides and air conditioning, it looked too much like keeping a baby in an aquarium to catch on.

In 1945, he became the chairman of the psychology department at Indiana University. In 1948, he was invited to come to Harvard, where he remained for the rest of his life. He was a very active man, doing research and guiding hundreds of doctoral candidates as well as writing many books. While not successful as a writer of fiction and poetry, he became one of our best psychology writers, including the book ***Walden II***, which is a fictional account of a community run by his behaviorist principles.

August 18, 1990, B. F. Skinner died of leukemia after becoming perhaps the most celebrated psychologist since Sigmund Freud.

Theory



B. F. Skinner’s entire system is based on **operant conditioning**. The organism is in the process of “operating” on the environment, which in ordinary terms means it is bouncing around its world, doing what it does. During this “operating,” the organism encounters a special kind of stimulus, called a **reinforcing stimulus**, or simply a reinforcer. This special stimulus has the effect of increasing the **operant** -- that is, the behavior occurring just before the reinforcer. This is operant conditioning: “the behavior is followed by a consequence, and the nature of the consequence modifies the organism's tendency to repeat the behavior in the future.”

Imagine a rat in a cage. This is a special cage (called, in fact, a “Skinner box”) that has a bar or pedal on one wall that, when pressed, causes a little mechanism to release a food pellet into the cage. The rat is bouncing around the cage, doing whatever it is rats do, when he accidentally presses the bar and -- hey, presto! -- a food pellet falls into the cage! The operant is the behavior just prior to the reinforcer, which is the food pellet, of course. In no time at all, the rat is furiously pedaling away at the bar, hoarding his pile of pellets in the corner of the cage.

A behavior followed by a reinforcing stimulus results in an increased probability of that behavior occurring in the future.

What if you don’t give the rat any more pellets? Apparently, he’s no fool, and after a few futile attempts, he stops his bar-pressing behavior. This is called **extinction** of the operant behavior.

A behavior no longer followed by the reinforcing stimulus results in a decreased probability of that behavior occurring in the future.

Now, if you were to turn the pellet machine back on, so that pressing the bar again provides the rat with pellets, the behavior of bar-pushing will “pop” right back into existence, much more quickly than it took for the rat to learn the behavior the first time. This is because the return of the reinforcer takes place in the context of a reinforcement history that goes all the way back to the very first time the rat was reinforced for pushing on the bar!

Schedules of reinforcement

Skinner likes to tell about how he "accidentally" -- i.e. operantly -- came across his various discoveries. For example, he talks about running low on food pellets in the middle of a study. Now, these were the days before “Purina rat chow” and the like, so Skinner had to make his own rat pellets, a slow and tedious task. So he decided to reduce the number of reinforcements he gave his rats for whatever behavior he was trying to condition, and, lo and behold, the rats kept up their operant behaviors, and at a stable rate, no less. This is how Skinner discovered **schedules of reinforcement**!

Continuous reinforcement is the original scenario: Every time that the rat does the behavior (such as pedal-pushing), he gets a rat goodie.

The **fixed ratio schedule** was the first one Skinner discovered: If the rat presses the pedal three times, say, he gets a goodie. Or five times. Or twenty times. Or “x” times. There is a fixed ratio between behaviors and reinforcers: 3 to 1, 5 to 1, 20 to 1, etc. This is a little like “piece rate” in the clothing manufacturing industry: You get paid so much for so many shirts.

The **fixed interval schedule** uses a timing device of some sort. If the rat presses the bar at least once during a particular stretch of time (say 20 seconds), then he gets a goodie. If he fails to do so, he doesn’t get a goodie. But even if he hits that bar a hundred times during that 20 seconds, he still only gets one goodie! One strange thing that happens is that the rats tend to “pace” themselves: They slow down the rate of their behavior right after the reinforcer, and speed up when the time for it gets close.

Skinner also looked at **variable schedules**. Variable ratio means you change the “x” each time -- first it takes 3 presses to get a goodie, then 10, then 1, then 7 and so on. Variable interval means you keep changing the time period -- first 20 seconds, then 5, then 35, then 10 and so on.

In both cases, it keeps the rats on their rat toes. With the variable interval schedule, they no longer “pace” themselves, because they no can no longer establish a “rhythm” between behavior and reward. Most importantly, these schedules are very resistant to extinction. It makes sense, if you think about it. If you haven’t gotten a reinforcer for a while, well, it could just be that you are at a particularly “bad” ratio or interval! Just one more bar press, maybe this’ll be the one!

This, according to Skinner, is the mechanism of gambling. You may not win very often, but you never know whether and when you’ll win again. It could be the very next time, and if you don’t roll them dice, or play that hand, or bet on that number this once, you’ll miss on the score of the century!

Shaping

A question Skinner had to deal with was how we get to more complex sorts of behaviors. He responded with the idea of **shaping**, or "the method of successive approximations." Basically, it involves first reinforcing a behavior only vaguely similar to the one desired. Once that is established, you look out for variations that come a little closer to what you want, and so on, until you have the animal performing a behavior that would never show up in ordinary life. Skinner and his students have been quite successful in teaching simple animals to do some quite extraordinary things. My favorite is teaching pigeons to bowl!

I used shaping on one of my daughters once. She was about three or four years old, and was afraid to go down a particular slide. So I picked her up, put her at the end of the slide, asked if she was okay and if she could jump down. She did, of course, and I showered her with praise. I then picked her up and put her a foot or so up the slide, asked her if she was okay, and asked her to slide down and jump off. So far so good. I repeated this again and again, each time moving her a little up the slide, and backing off if she got nervous. Eventually, I could put her at the top of the slide and she could slide all the way down and jump off. Unfortunately, she still couldn't climb up the ladder, so I was a very busy father for a while.

Beyond these fairly simple examples, shaping also accounts for the most complex of behaviors. You don't, for example, become a brain surgeon by stumbling into an operating theater, cutting open someone's head, successfully removing a tumor, and being rewarded with prestige and a hefty paycheck, along the lines of the rat in the Skinner box. Instead, you are gently shaped by your environment to enjoy certain things, do well in school, take a certain bio class, see a doctor movie perhaps, have a good hospital visit, enter med school, be encouraged to drift towards brain surgery as a speciality, and so on. This could be something your parents were carefully doing to you, *à la* a rat in a cage. But much more likely, this is something that was more or less unintentional.

Aversive stimuli

An **aversive stimulus** is the opposite of a reinforcing stimulus, something we might find unpleasant or painful.

A behavior followed by an aversive stimulus results in a decreased probability of the behavior occurring in the future.

This both defines an aversive stimulus and describes the form of conditioning known as **punishment**. If you shock a rat for doing x, it'll do a lot less of x. If you spank Johnny for throwing his toys he will throw his toys less and less (maybe).

On the other hand, if you remove an already active aversive stimulus after a rat or Johnny performs a certain behavior, you are doing **negative reinforcement**. If you turn off the electricity when the rat stands on his hind legs, he'll do a lot more standing. If you stop your perpetual nagging when I finally take out the garbage, I'll be more likely to take out the garbage (perhaps). You could say it "feels so good" when the aversive stimulus stops, that this serves as a reinforcer!

Behavior followed by the removal of an aversive stimulus results in an increased probability of that behavior occurring in the future.

Notice how difficult it can be to distinguish some forms of negative reinforcement from positive reinforcement: If I starve you, is the food I give you when you do what I want a positive -- i.e. a reinforcer? Or is it the removal of a negative -- i.e. the aversive stimulus of hunger?

Skinner (contrary to some stereotypes that have arisen about behaviorists) doesn't "approve" of the use of aversive stimuli -- not because of ethics, but because they don't work well! Notice that I said earlier that Johnny will maybe stop throwing his toys, and that I perhaps will take out the garbage? That's because whatever was reinforcing the bad behaviors hasn't been removed, as it would've been in the case of extinction. This hidden reinforcer has just been "covered up" with a conflicting aversive stimulus. So, sure, sometimes the child (or me) will behave -- but it still feels good to throw those toys. All Johnny needs to do is wait till you're out of the room, or find a way to blame it on his brother, or in some way escape the consequences, and he's back to his old ways. In fact, because Johnny now only gets to enjoy his reinforcer occasionally, he's gone into a variable schedule of reinforcement, and he'll be even more resistant to extinction than ever!

Behavior modification

Behavior modification -- often referred to as **b-mod** -- is the therapy technique based on Skinner's work. It is very straightforward: Extinguish an undesirable behavior (by removing the reinforcer) and replace it with a desirable behavior by reinforcement. It has been used on all sorts of psychological problems -- addictions, neuroses, shyness, autism, even schizophrenia -- and works particularly well with children. There are examples of back-ward psychotics who haven't communicated with others for years who have been conditioning to behave themselves in fairly normal ways, such as eating with a knife and fork, taking care of their own hygiene needs, dressing themselves, and so on.

There is an offshoot of b-mod called the **token economy**. This is used primarily in institutions such as psychiatric hospitals, juvenile halls, and prisons. Certain rules are made explicit in the institution, and behaving yourself appropriately is rewarded with tokens -- poker chips, tickets, funny money, recorded notes, etc. Certain poor behavior is also often followed by a withdrawal of these tokens. The tokens can be traded in for desirable things such as candy, cigarettes, games, movies, time out of the institution, and so on. This has been found to be very effective in maintaining order in these often difficult institutions.

There is a drawback to token economy: When an "inmate" of one of these institutions leaves, they return to an environment that reinforces the kinds of behaviors that got them into the institution in the first place. The psychotic's family may be thoroughly dysfunctional. The juvenile offender may go right back to "the 'hood." No one is giving them tokens for eating politely. The only reinforcements may be attention for "acting out," or some gang glory for robbing a Seven-Eleven. In other words, the environment doesn't travel well!

Walden II

Skinner started his career as an English major, writing poems and short stories. He has, of course, written a large number of papers and books on behaviorism. But he will probably be most remembered by the general run of readers for his book ***Walden II***, wherein he describes a utopia-like commune run on his operant principles.

People, especially the religious right, came down hard on his book. They said that his ideas take away our freedom and dignity as human beings. He responded to the sea of criticism with another book (one of his best) called ***Beyond Freedom and Dignity***. He asked: What do we mean when we say we want to be free? Usually we mean we don't want to be in a society that punishes us for doing what we want to do. Okay -- aversive stimuli don't work well anyway, so out with them! Instead, we'll only use reinforcers to "control" society. And if we pick the right reinforcers, we will feel free, because we will be doing what we feel we want!

Likewise for dignity. When we say "she died with dignity," what do we mean? We mean she kept up her "good" behaviors without any apparent ulterior motives. In fact, she kept her dignity because her reinforcement history has led her to see behaving in that "dignified" manner as more reinforcing than making a scene.

The bad do bad because the bad is rewarded. The good do good because the good is rewarded. There is no true freedom or dignity. Right now, our reinforcers for good and bad behavior are chaotic and out of our control -- it's a matter of having good or bad luck with your "choice" of parents, teachers, peers, and other influences. Let's instead take control, as a society, and design our culture in such a way that good gets rewarded and bad gets extinguished! With the right **behavioral technology**, we can **design culture**.

Both freedom and dignity are examples of what Skinner calls **mentalistic constructs** -- unobservable and so useless for a scientific psychology. Other examples include defense mechanisms, the unconscious, archetypes, fictional finalisms, coping strategies, self-actualization, consciousness, even things like hunger and thirst. The most important example is what he refers to as the **homunculus** -- Latin for "the little man" -- that supposedly resides inside us and is used to explain our behavior, ideas like soul, mind, ego, will, self, and, of course, personality.

Instead, Skinner recommends that psychologists concentrate on observables, that is, the environment and our behavior in it.

Skinner was to enjoy considerable popularity during the 1960's and even into the 70's. But both the humanistic movement in the clinical world, and the cognitive movement in the experimental world, were quickly moving in on his beloved behaviorism. Before his death, he publicly lamented the fact that the world had failed to learn from him.

5.2: Walden Two

Walden Two

[A Selection]

B. F. Skinner

Chapter 13

The quarters for children from one to I three consisted of several small playrooms with Lilliputian furniture, a child's lavatory, and a dressing and locker room. Several small sleeping rooms were operated on the same principle as the baby cubicles. The temperature and the humidity were controlled so that clothes or bedclothing were not needed. The cots were double-decker arrangements of the plastic mattresses we had seen in the cubicles. The children slept unclothed, except for diapers. There were more beds than necessary, so that the children could be grouped according to developmental age or exposure to contagious diseases or need for supervision, or for educational purposes.

We followed Mrs. Nash to a large screened porch on the south side of the building, where several children were playing in sandboxes and on swings and climbing apparatuses. A few wore "training pants"; the rest were naked. Beyond the porch was a grassy play yard enclosed by closely trimmed hedges, where other children, similarly undressed, were at play. Some kind of marching game was in progress.

As we returned, we met two women carrying food hampers. They spoke to Mrs. Nash and followed her to the porch. In a moment five or six children came running into the playrooms and were soon using the lavatory and dressing themselves. Mrs. Nash explained that they were being taken on a picnic.

"What about the children who don't go?" said Castle. "What do you do about the green-eyed monster?"

Mrs. Nash was puzzled.

"Jealousy. Envy," Castle elaborated. "Don't the children who stay home ever feel unhappy about it?"

"I don't understand," said Mrs. Nash.

"And I hope you won't try," said Frazier with a smile. "I'm afraid we must be moving along."

We said good-bye, and I made an effort to thank Mrs. Nash, but she seemed to be puzzled by that too, and Frazier frowned as if I had committed some breach of good taste.

"I think Mrs. Nash's puzzlement?" said Frazier, as we left the building, "is proof enough that our children are seldom envious or jealous. Mrs. Nash was twelve years old when Walden Two was founded. It was a little late to undo her early training, but I think we were successful. She's a good example of the Walden Two product. She could probably recall the experience of jealousy, but it's not part of her present life."

"Surely that's going too far!" said Castle. "You can't be so godlike as all that! You must be assailed by emotions just as much as the rest of us!"

“We can discuss the question of godlikeness later, if you wish," replied Frazier. "As to emotions—we aren't free of them all, nor should we like to be. But the meaner and more annoying—the emotions which breed unhappiness--are almost unknown here, like unhappiness itself. We don't need them any longer in our struggle for existence, and it's easier on our circulatory system, and certainly pleasantry, to dispense with them.”

"If you've discovered how to do that. you are indeed a genius," said Castle. He seemed almost stunned as Frazier nodded assent. "We all know that emotions are useless and bad for our peace of mind and our blood pressure ' he went on. "But how arrange things otherwise?"

“We arrange them otherwise here," said Frazier. He was showing a mildness of manner which I was coming to recognize as a sign of confidence.

"But emotions are—fun!" said Barbara. "Life wouldn't be worth living without them."

"Some of them, yes” said Frasier. “The productive and strengthening emotions—joy and love. But sorrow and hate—and the high-voltage excitements of anger, fear, and rage are out of proportion with the needs of modern life, and they're wasteful and dangerous. Mr. Castle has mentioned jealousy, a minor form of anger, I think we may call it. Naturally we avoid it. It has served its purpose in the evolution of man; we've no further use for it. If we allowed it to persist, it would only sap the life out of us. In a cooperative society there's no jealousy because there's no need for jealousy."

"That implies that you all get everything you want," said Castle. "But what about social possessions? Last night you mentioned the young man who chose a particular girl or profession. There's still a chance for jealousy there, isn't there?"

"It doesn't imply that we get everything we want," said Frazier. "Of course we don't. But jealousy wouldn't help. In a competitive world there's some point to it. It energizes one to attack a frustrating condition. The impulse and the added energy are an advantage. Indeed, in a competitive world emotions work all too well. Look at the singular lack of success of the complacent man. He enjoys a more serene life, but it's less likely to be a fruitful one. The world isn't ready for simple pacifism or Christian humility, to cite two cases in point. Before you can safely turn out the destructive and wasteful emotions, you must make sure they're no longer needed."

"How do you make sure that jealousy isn't needed in Walden Two?" I said.

"In Walden Two problems can't be solved by attacking others" said Frazier with marked finality.

"That's not the same as eliminating jealousy, though" I said.

"Of course it's not. But when a particular emotion is no longer a useful part of a behavioral repertoire, we proceed to eliminate it."

"Yes, but how?"

"It's simply a matter of behavioral engineering," said Frazier.

"Behavioral engineering?"

"You're baiting me, Burris. You know perfectly well what I mean. The techniques have been available for centuries. We use them in education and in the psychological management of the community. But you're forcing my hand" he added. "I was saving that for this evening. But let's strike while the iron is hot."

We had stopped at the door of the large children's building. Frazier shrugged his shoulders, walked to the shade of a large tree, and threw himself on the ground. We arranged ourselves about him and waited.

Chapter 14

Each of us," Frazier began, "is engaged in a pitched battle with the rest of mankind."

"A curious premise for a Utopia," said Castle. "Even a pessimist like myself takes a more hopeful view than that."

"You do, you do," said Frazier. "But lets be realistic. Each of us has interests which conflict with the interests of everybody else. That's our original sin, and it can't be helped. Now, 'everybody else' we call 'society.' It's a powerful opponent, and it always wins. Oh, here and there an individual prevails for a while and gets what he wants. Sometimes he storms the culture of a society and changes it slightly to his own advantage. But society wins in the long run, for it has the advantage of numbers and of age. Many prevail against one, and men against a baby. Society attacks early, when the individual is helpless. It enslaves him almost before he has tasted freedom. The 'ologies' will tell you how its done. Theology calls it building a conscience or developing a spirit of selfless. Psychology calls it the growth of the super ego.

"Considering how long society has been at it, you'd expect a better job. But the campaigns have been badly planned and the victory has never been secure. The behavior of the individual has been shaped according to revelations of 'good conduct,' never as the result of experimental study. But why not experiment? The questions are simple enough. What's the best behavior for the individual so far as the group is concerned? And how can the individual be induced to behave in that way? Why not explore these questions in a scientific spirit?

"We could do just that in Walden Two. We had already worked out a code of conduct—subject, of course, to experimental modification. The code would keep things running smoothly if everybody lived up to it. Our job was to see that everybody did. Now, you can't get people to follow a useful code by making them into so many jack-in-the-boxes. You can't foresee all future circumstances, and you can't specify adequate future conduct. You don't know what will be required. Instead you have to set up certain behavioral processes which lead the individual to design his own 'good' conduct when the time comes. We call that sort of thing 'self-control.' But don't be misled, the control always rests in the last analysis in the hands of society.

"One of our Planners, a young man named Simmons, worked with me. It was the first time in history that the matter was approached in an experimental way. Do you question that statement, Mr. Castle?"

“I'm not sure I know what you are talking about," said Castle.

“Then let me go on. Simmons and I began by studying the great works on morals and ethics—Plato, Aristotle, Confucius, the New Testament, the Puritan divines, Machiavelli, Chesterfield, Freud—there were scores of them. We were looking for any and every method of shaping human behavior by imparting techniques of self-control. Some techniques were obvious enough, for they had marked turning points in human history. 'Love your enemies' is an example--a psychological invention for easing the lot of an oppressed people. The severest trial of oppression is the constant rage which one suffers at the thought of the oppressor. What Jesus discovered was how to avoid these inner devastations. His technique was to practice the opposite emotion. If a man can succeed in loving his enemies and 'taking no thought for the morrow,' he will no longer be assailed by hatred of the oppressor or rage at the loss of his freedom or possessions. He may not get his freedom or possessions back, but he's less miserable. It's a difficult lesson. It comes late in our program."

"I thought you were opposed to modifying emotions and instinct until the world was ready for it," said Castle. "According to you, the principle of love your enemies' should have been suicidal."

"It would have been suicidal, except for an entirely unforeseen consequence. Jesus must have been quite astonished at the effect of his discovery. We are only just beginning to understand the power of love because we are just beginning to understand the weakness of force and aggression. But the science of behavior is clear about all that now. Recent discoveries in the analysis of punishment—but I am falling into one digression after another. Let me save my explanation of why the Christian virtues—and I mean merely the Christian techniques of self-control—have not disappeared from the face of the earth, with due recognition of the fact that they suffered a narrow squeak within recent memory.

"When Simmons and I had collected our techniques of control, we had to discover how to teach them. That was more difficult. Current educational practices were of little value, and religious practices scarcely any better. Promising paradise or threatening hell-fire is, we assumed, generally admitted to be unproductive. It is based upon a fundamental fraud which, when discovered, turns the individual against society and nourishes the very thing it tries to stamp out. What Jesus offered in return for loving one's enemies was heaven on earth, better known as peace of mind.

"We found a few suggestions worth following in the practices of the clinical psychologist. We undertook to build a tolerance for annoying experiences. The sun shine of midday is extremely painful if you come from a dark room, but take it in easy stages and you can avoid pain altogether. The analogy can be misleading, but in much the same way it's possible to build a tolerance to painful or distasteful stimuli, or to frustration, or to situations which arouse fear, anger or rage. Society and nature throw these annoyances at the individual with no regard for the development of tolerances. Some achieve tolerances, most fail. Where would the science of immunization be if it followed a schedule of accidental dosages?

“Take the principle of 'Get thee behind me, Satan,' for example," Frazier continued. "It's a special case of self-control by altering the environment. Subclass A 3, I believe. We give each child a lollipop which has been dipped in powdered sugar so that a single touch of the tongue can be detected. We tell him he may eat the lollipop later in the day, provided it hasn't already been licked. Since the child is only three or four, it is a fairly diff---- "

"Three or four!" Castle exclaimed.

"All our ethical training is completed by the age of six," said Frazier quietly. "A simple principle like putting temptation out of sight would be acquired before four. But at such an early age the problem of not licking the lollipop isn't easy. Now, what would you do, Mr. Castle, in a similar situation?"

"Put the lollipop out of sight as quickly as possible."

"Exactly. I can see you've been well trained. Or perhaps you discovered the principle for yourself. We're in favor of original inquiry wherever possible, but in this case we have a more important goal and we don't
hesitate to give verbal help. First of all, the children are urged to examine their own behavior while looking at the lollipops. This helps them to recognize the need for self-control. Then the lollipops are concealed, and the children are asked to notice any gain in happiness or any reduction in tension. Then a strong distraction is arranged—say, an interesting game. Later the children are reminded of the candy and encouraged to examine their reaction. The value of the distraction is generally obvious. Well, need I go on? When the experiment is repeated a day or so later, the children all run with the lollipops to their lockers and do exactly what Mr. Castle would do—a sufficient indication of the success of our training."

"I wish to report an objective observation of my reaction to your story," said Castle, controlling his voice with great precision. "I find myself revolted by this display of sadistic tyranny."

"I don't wish to deny you the exercise of an emotion which you seem to find enjoyable," said Frazier. "So let me go on. Concealing a tempting but forbidden object is a crude solution. For one thing, it's not always feasible. We want a sort of psychological concealment—covering up the candy by paying no attention. In a later experiment the children wear their lollipops like crucifixes for a few hours."

" 'Instead of the cross, the lollipop,
About my neck was hung,' "
said Castle.

"I wish somebody had taught me that, though," said Rodge, with a glance at Barbara.

"Don't we all?" said Frazier. "Some of us learn control, more or less by accident. The rest of us go all our lives not even understanding how it is possible, and blaming our failure on being born the wrong way."

"How do you build up a tolerance to an annoying situation?" I said.

"Oh, for example, by having the children 'take' a more and more painful shock, or drink cocoa with less and less sugar in it until a bitter concoction can be savored without a bitter face."

"But jealousy or envy—you can't administer them in graded doses," I said.

"And why not? Remember, we control the social environment, too, at this age. That's why we get our ethical training in early. Take this case. A group of children arrive home after a long walk tired and hungry. They're expecting supper; they find, instead, that it's time for a lesson in self-control: they must stand for five minutes in front of steaming bowls of soup.

"The assignment is accepted like a problem in arithmetic. Any groaning or complaining is a wrong answer. Instead, the children begin at once to work upon themselves to avoid any unhappiness during the delay. One of them may make a joke of it. We encourage a sense of humor as a good way of not taking an annoyance seriously. The joke won't be much, according to adult standards—perhaps the child will simply pretend to empty the bowl of soup into his upturned mouth. Another may start a song with many verses. The rest join in at once, for they've learned that it's a good way to make time pass."

Frazier glanced uneasily at Castle, who was not to be appeased.

“That also strikes you as a form of torture, Mr. Castle?" he asked.

"I'd rather be put on the rack," said Castle.

"Then you have by no means had the thorough training I supposed. You can’t imagine how lightly the children take such an experience. It's a rather severe biological frustration, for the children are tired and hungry and they must stand and look at food; but it's passed off as lightly as a five-minute delay at curtain time. We regard it as a fairly elementary test. Much more difficult problems follow."

"I suspected as much," muttered Castle.

"In a later stage we forbid all social devices. No songs, no jokes—merely silence. Each child is forced back upon his own resources—a very important step."

"I should think so," I said. "And how do you know it's successful? You might produce a lot of silently resentful children. It's certainly a dangerous stage."

"It is, and we follow each child carefully. If he hasn't picked up the necessary techniques, we start back a little. A still more advanced stage"—Frazier glanced again at Castle, who stirred uneasily—"brings me to my point. When it's time to sit down to the soup, the children count off—heads and tails. Then a coin is tossed and if it comes up heads, the 'heads' sit down and eat. The 'tails' remain standing for another five minutes."

Castle groaned.

"And you call that envy?" I asked.

"Perhaps not exactly," said Frazier. "At least there's seldom any aggression against the lucky ones. The emotion, if any, is directed against Lady Luck herself, against the toss of the coin. That, in itself, is a lesson worth learning, for it's the only direction in which emotion has a surviving chance to be useful. And resentment toward things in general, while perhaps just as silly as personal aggression, is more easily controlled. Its expression is not socially objectionable."

Frazier looked nervously from one of us to the other. He seemed to be trying to discover whether we shared Castle's prejudice. I began to realize, also, that he had not really wanted to tell this story. He was vulnerable. He was treading on sanctified ground, and I was pretty sure he had not established the value of most of these practices in an experimental fashion. He could scarcely have done so in the short space of ten years. He was working on faith, and it bothered him.

I tried to bolster his confidence by reminding him that he had a professional colleague among his listeners. "May you not inadvertently teach your children some of the very emotions you're trying to eliminate?" I said. "What's the effect, for example, of finding the anticipation of a warm supper suddenly thwarted? Doesn't that eventually lead to feelings of uncertainty, or even anxiety?"

"It might. We had to discover how often our lessons could be safely administered. But all our schedules are worked out experimentally. We watch for undesired consequences just as any scientist watches for disrupting factors in his experiments.

"After all, it's a simple and sensible program," he went on in a tone of appeasement. "We set up a system of gradually increasing annoyances and frustrations against a background of complete serenity. An easy environment is made more and more difficult as the children acquire the capacity to adjust."

"But why?" said Castle. "Why these deliberate unpleasantnesses—to put it mildly? I must say I think you and your friend Simmons are really very subtle sadists."

“You've reversed your position, Mr. Castle," said Frazier in a sudden flash of anger with which I rather sympathized. Castle was calling names, and he was also being unaccountably and perhaps intentionally obtuse. "A while ago you accused me of breeding a race of softies," Frazier continued. "Now you object to toughening them up. But what you don't understand is that these potentially unhappy situations are never very annoying. Our schedules make sure of that. You wouldn't understand, however, because you're not so far advanced as our children."

Castle grew black.

"But what do your children get out of it?" he insisted, apparently trying to press some vague advantage in Frazier's anger.

"What do they get out of it!" exclaimed Frazier, his eyes flashing with a sort of helpless contempt. His lips curled and he dropped his head to look at his fingers, which were crushing a few blades of grass.

"They must get happiness and freedom and strength," I said, putting myself in a ridiculous position in attempting to make peace.

"They don't sound happy or free to me, standing in front of bowls of Forbidden Soup," said Castle, answering me parenthetically while continuing to stare at Frazier.

"If I must spell it out," Frazier began with a deep sigh, "what they get is escape from the petty emotions which eat the heart out of the unprepared. They get the satisfaction of pleasant and profitable social relations on a scale almost undreamed of in the world at large. They get immeasurably increased efficiency, because they can stick to a job without suffering the aches and pains which soon beset most of us. They get new horizons, for they are spared the emotions characteristic of frustration and failure. They get—"His eyes searched the branches of the trees. "Is that enough?,” he said at last.

"And the community must gain their loyalty," I said, "when they discover the fears and jealousies and diffidences in the world at large."

"I'm glad you put it that way," said Frazier. “You might have said that they must feel superior to the miserable products of our public schools. But we're at pains to keep any feeling of superiority or contempt under control, too. Having suffered most acutely from it myself, I put the subject first on our agenda. We carefully avoid any joy in a personal triumph which means the personal failure of somebody else. We take no pleasure in the sophistical, the disputative, the dialectical." He threw a vicious glance at Castle. "We don't use the motive of domination, because we are always thinking of the whole group. We could motivate a few geniuses that way—it was certainly my own motivation—but we'd sacrifice some of the happiness of everyone else. Triumph over nature and over oneself, yes. But over others, never."

"You've taken the mainspring out of the watch," said Castle flatly.

"That's an experimental question, Mr. Castle, and you have the wrong answer."

Frazier was making no effort to conceal his feeling. If he had been riding Castle, he was now using his spurs. Perhaps he sensed that the rest of us had come round and that he could change his tactics with a single holdout. But it was more than strategy, it was genuine feeling. Castle's undeviating skepticism was a growing frustration.

"Are your techniques really so very new?" I said hurriedly. "What about the primitive practice of submitting a boy to various tortures before granting him a place among adults? What about the disciplinary techniques of Puritanism? Or of the modern school, for that matter?"

"In one sense you're right," said Frazier. "And I think you've nicely answered Mr. Castle's tender concern for our little ones. The unhappinesses we deliberately impose are far milder than the normal unhappinesses from which we offer protection. Even at the height of our ethical training, the unhappiness is ridiculously trivial—to the well-trained child.

"But there's a world of difference in the way we use these annoyances," he continued. "For one thing, we don't punish. We never administer an unpleasantness in the hope of repressing or eliminating undesirable behavior. But there's another difference. In most cultures the child meets up with annoyances and reverses of uncontrolled magnitude. Some are imposed in the name of discipline by persons in authority. Some, like hazings, are condoned though not authorized. Others are merely accidental. No one cares to, or is able to, prevent them.

"We all know what happens. A few hardy children emerge, particularly those who have got their unhappiness in doses that could be swallowed. They become brave men. Others become sadists or masochists of varying degrees of pathology. Not having conquered a painful environment, they become preoccupied with pain and make a devious art of it. Others submit—and hope to inherit the earth. The rest—the cravens, the cowards—live in fear for the rest of their lives. And that's only a single field—the reaction to pain. I could cite a dozen parallel cases. The optimist and the pessimist, the contented and the disgruntled, the loved and the unloved, the ambitious and the discouraged— these are only the extreme products of a miserable system.

"Traditional practices are admittedly better than nothing," Frazier went on. "Spartan or Puritan—no one can question the occasional happy result. But the whole system rests upon the wasteful principle of selection. The English public school of the nineteenth century produced brave men—by setting up almost insurmountable barriers and making the most of the few who came over. But selection isn't education. Its crops of brave men will always be small, and the waste enormous. Like all primitive principles, selection serves in place of education only through a profligate use of material. Multiply extravagantly and select with rigor. Its the philosophy of the 'big litter' as an alternative to good child hygiene.

"In Walden two we have a different objective. We make every man a brave man. They all come over the barriers. Some require more preparation than others, but they all come over. The traditional use of adversity is to select the strong. We control adversity to build strength. And we do it deliberately, no matter how sadistic Mr. Castle may think us, in order to prepare for adversities which are beyond control. Our children eventually experience the 'heartache and the thousand natural shocks that flesh is heir to.' It would be the cruelest possible practice to protect them as long as possible, especially when we could protect them so well."

Frazier held out his hands in an exaggerated gesture of appeal.

"What alternative had we?" he said, as if he were in pain. "What else could we do? For four or five years we could provide a life in which no important need would go unsatisfied, a life practically free of anxiety or frustration or annoyance. What would you do? Would you let the child enjoy this paradise with no thought for the future—like an idolatrous and pampering mother? Or would you relax control of the environment and let the child meet accidental frustrations? But what is the virtue of accident? No, there was only one course open to us. We had to design a series of adversities, so that the child would develop the greatest possible self-control. Call it deliberate, if you like, and accuse us of sadism; there was no other course." Frazier turned to Castle, but he was scarcely challenging him. He seemed to be waiting, anxiously, for his capitulation. But Castle merely shifted his ground.

"I find it difficult to classify these practices," he said. Frazier emitted a disgruntled "Ha!" and sat back. “Your system seems to have usurped the place as well as the techniques of religion."

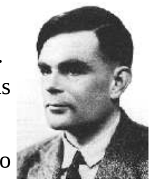
"Of religion and family culture," said Frazier wearily. "But I don't call it usurpation. Ethical training belongs to the community. As for techniques, we took every suggestion we could find without prejudice as to the source. But not on faith. We disregarded all claims of revealed truth and put every principle to an experimental test. And by the way, I've very much misrepresented the whole system if you suppose that any of the practices I've described are fixed. We try out many different techniques. Gradually we work toward the best possible set. And we don't pay much attention to the apparent success of a principle in the course of history. History is honored in Walden Two only as entertainment. It isn't taken seriously as food for thought. Which reminds me, very rudely, of our original plan for the morning. Have you had enough of emotion? Shall we turn to intellect?"

Frazier addressed these questions to Castle in a very friendly way and I was glad to see that Castle responded in kind. It was perfectly clear, however, that neither of them had ever worn a lollipop about the neck or faced a bowl of Forbidden Soup.

5.3: The Cognitive Movement

In the latter half of the twentieth century, the advent of the computer and the way of thinking associated with it led to a new approach or orientation to psychology called **the cognitive movement**. Many are hoping that it will prove to be the paradigm -- the unifying theory -- we have been waiting for. It is still way too early to tell, but the significance of cognitive psychology is impossible to deny.

Alan M. Turing



Alan Turing was born June 23, 1912 in Paddington, London, the second child of Julius Mathison Turing and Ethel Sara Stoney. His parents met while his father and his mother's father were serving in Madras, India, as part of the Civil Service. He and his brother were raised in other people's homes while his parents continued their life in India.

A turning point in his life came when his best friend at Sherborne School, Christopher Marcom, died in 1930. This led him to think about the nature of existence and whether or not it ends at death.

He went to King's College of Cambridge in 1931, where he read books by von Neumann, Russell and Whitehead, Goedel, and so on. He also became involved in the pacifist movement at Cambridge, as well as coming to terms with his homosexuality. He received his degree in 1934, and stayed on for a fellowship in 1935.

The **Turing Machine** -- the first description of what would become the modern computer -- was introduced in a 1936 paper, after which he left for Princeton in the US. There, he received his PhD in 1938, and returned to King's College, living on his fellowship.

He began working with British Intelligence on breaking the famous Enigma Code by constructing code-breaking machines. In 1944, he made his first mention of "building a brain."

It should be noted that Turing was also an amateur cross-country runner, and just missed representing the UK in the 1948 Olympics!

In 1944, he became the deputy director of the computing lab at Manchester University, where they were attempting to build the first true computer. In 1950, he published a paper, "Computing Machinery and Intelligence," in **Mind**. Turing saw the human brain as an "unorganized machine" that learned through experience.

Unfortunately, he was arrested and tried in 1952 -- for homosexuality! He made no defense, but took an offer to stay out of jail if he would take estrogen injections to lower his supposedly overactive libido. He lost security clearance because of his homosexuality as well.

He began working on pattern formation in biology -- what we would now call the mathematics of fractals -- and on quantum mechanics. But on June 7, 1954, he committed suicide by ingesting cyanide -- making it look like an accident to spare his mother's feelings. He was 41.

Today, he is considered the father of Computer Science. Let me let his biographer, Andrew Hodges, describe the famous Turing Machine:

> *His work introduced a concept of immense practical significance: the idea of the Universal Turing Machine. The concept of 'the Turing machine' is like that of 'the formula' or 'the equation'; there is an infinity of possible Turing machines, each corresponding to a different 'definite method' or algorithm. But imagine, as Turing did, each particular algorithm written out as a set of instructions in a standard form. Then the work of interpreting the instructions and carrying them out is itself a mechanical process, and so can itself be embodied in a particular Turing machine, namely the Universal Turing machine. A Universal Turing machine can be made do what any other particular Turing machine would do, by supplying it with the standard form describing that Turing machine. One machine, for all possible tasks.*
>
> *It is hard now not to think of a Turing machine as a computer program, and the mechanical task of interpreting and obeying the program as what the computer itself does. Thus, the Universal Turing Machine embodies the essential principle of the computer: a single machine which can be turned to any well-defined task by being supplied with the appropriate program.*
>
> *Additionally, the abstract Universal Turing Machine naturally exploits what was later seen as the 'stored program' concept essential to the modern computer: it embodies the crucial twentieth-century insight that symbols representing instructions are no different in kind from symbols representing numbers. But computers, in this modern sense, did not exist in 1936. Turing created*

> *these concepts out of his mathematical imagination. Only nine years later would electronic technology be tried and tested sufficiently to make it practical to transfer the logic of his ideas into actual engineering. In the meanwhile the idea lived only in his mind.*

Quoted from Andrew Hodges' "Alan Turing --- a short biography," at http://www.turing.org.uk/turing/bio/part3.html

For much more on Turing, see the Turing Archive at http://www.cs.usfca.edu/www.AlanTuring.net/turing_archive/index.html

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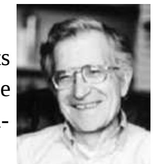
Noam Chomsky



In addition to the input (no pun intended) from the "artificial intelligence" people, there was the input from a group of scientists in a variety of fields who thought of themselves as structuralists -- not allying themselves with Wundt, but interested in the structure of their various topics. I'll call them neo-structuralists, just to keep them straight. For example, there's Claude Levi-Strauss, the famous French anthropologist. But the one everyone knows about is the linguist Noam Chomsky.

Avram Noam Chomsky was born December 7, 1928, in Philadelphia, the son of William Chomsky and Elsie Simonofsky. He was father was a Hebrew scholar, and young Noam became so good that he was proof-reading his father's manuscripts by the time he was in high school. Noam was also passionate about politics, especially when it concerned the potential for a state of Israel.

He received his BA from the University of Pennsylvania in 1949, whereupon he married a fellow linguist, Carol Schatz. They would go on to have three children. He received his PhD in 1955, also from the U of Penn.

That same year, he started teaching at MIT and began his work on **generative grammar**. Generative grammar was based on the question "how can we create new sentences which have never been spoken before?" How, in other words, do we get so creative, so generative? While considering this questions, he familiarized himself with mathematical logic, the psychology of thought, and theories about thinking machines. He found himself, on the other hand, very critical of traditional linguistics and behavioristic psychology.

In 1957, he published his first book, **Syntactic Structures**. Besides introducing his generative grammar, he also introduced the idea of an innate ability to learn languages. We have born into us a "universal grammar" ready to absorb the details of whatever language is presented to us at an early age.

His book spoke about **surface structure** and **deep structure** and the **rules of transformation** that governed the relations between them. Surface structure is essentially language as we know it, particular languages with particular rules of phonetics and basic grammar. Deep structure is more abstract, at the level of meanings and the universal grammar.

In the 1960's, Chomsky became one of the most vocal critics of the Vietnam War, and wrote American Power and the New Mandarins, a critique of government decision making. He is still at MIT today and continues to produce articles and books on linguistics -- and politics!

Jean Piaget

Another neo-structuralist is Jean Piaget. Originally a biologist, he is now best remembered for his work on the development of cognition. Many would argue that he, more than anyone else, is responsible for the creation of cognitive psychology. If the English-speaking world had only learned to read a little French, this would be true without a doubt. Unfortunately, his work was only introduced in English after 1950, and only became widely known in the 1960's -- just on time to be a part of the cognitive movement, but not of its creation.

Jean Piaget was born in Neuchâtel, Switzerland, on August 9, 1896. His father, Arthur Piaget, was a professor of medieval literature with an interest in local history. His mother, Rebecca Jackson, was intelligent and energetic, but Jean found her a bit neurotic -- an impression that he said led to his interest in psychology, but away from pathology! The oldest child, he was quite independent and took an early interest in nature, especially the collecting of shells. He published his first "paper"when he was ten -- a one page account of his sighting of an albino sparrow.

He began publishing in earnest in high school on his favorite subject, mollusks. He was particularly pleased to get a part time job with the director of Nuechâtel's Museum of Natural History, Mr. Godel. His work became well known among European students of mollusks, who assumed he was an adult! All this early experience with science kept him away, he says, from "the demon of philosophy."

Later in adolescence, he faced a bit a crisis of faith: Encouraged by his mother to attend religious instruction, he found religious argument childish. Studying various philosophers and the application of logic, he dedicated himself to finding a "biological explanation of knowledge." Ultimately, philosophy failed to assist him in his search, so he turned to psychology.

After high school, he went on to the University of Neuchâtel. Constantly studying and writing, he became sickly, and had to retire to the mountains for a year to recuperate. When he returned to Neuchâtel, he decided he would write down his philosophy. A fundamental point became a centerpiece for his entire life's work: "In all fields of life (organic, mental, social) there exist 'totalities' qualitatively distinct from their parts and imposing on them an organization." This principle forms the basis of his structuralist philosophy, as it would for the Gestaltists, Systems Theorists, and many others.

In 1918, Piaget received his Doctorate in Science from the University of Neuchâtel. He worked for a year at psychology labs in Zurich and at Bleuler's famous psychiatric clinic. During this period, he was introduced to the works of Freud, Jung, and others. In 1919, he taught psychology and philosophy at the Sorbonne in Paris. Here he met Simon (of Simon-Binet fame) and did research on intelligence testing. He didn't care for the "right-or-wrong" style of the intelligent tests and started interviewing his subjects at a boys school instead, using the psychiatric interviewing techniques he had learned the year before. In other words, he began asking how children reasoned.

In 1921, his first article on the psychology of intelligence was published in the **Journal de Psychologie**. In the same year, he accepted a position at the Institut J. J. Rousseau in Geneva. Here he began with his students to research the reasoning of elementary school children. This research became his first five books on child psychology. Although he considered this work highly preliminary, he was surprised by the strong positive public reaction to his work.

In 1923, he married one of his student coworkers, Valentine Châtenay. In 1925, their first daughter was born; in 1927, their second daughter was born; and in 1931, their only son was born. They immediately became the focus of intense observation by Piaget and his wife. This research became three more books!

In 1929, Piaget began work as the director of the Bureau International Office de l'Education, in collaboration with UNESCO. He also began large scale research with A. Szeminska, E. Meyer, and especially Bärbel Inhelder, who would become his major collaborator. Piaget, it should be noted, was particularly influential in bringing women into experimental psychology. Some of this work, however, wouldn't reach the world outside of Switzerland until World War II was over.

In 1940, He became chair of Experimental Psychology, the Director of the psychology laboratory, and the president of the Swiss Society of Psychology. In 1942, he gave a series of lectures at the Collège de France, during the Nazi occupation of France. These lectures became **The Psychology of Intelligence**. At the end of the war, he was named President of the Swiss Commission of UNESCO.

Also during this period, he received a number of honorary degrees. He received one from the Sorbonne in 1946, the University of Brussels and the University of Brazil in 1949, on top of an earlier one from Harvard in 1936. And, in 1949 and 1950, he published his synthesis, **Introduction to Genetic Epistemology**.

In 1952, he became a professor at the Sorbonne. In 1955, he created the International Center for Genetic Epistemology, of which he served as director the rest of his life. And, in 1956, he created the School of Sciences at the University of Geneva.

He continued working on a general theory of structures and tying his psychological work to biology for many more years. Likewise, he continued his public service through UNESCO as a Swiss delegate. By the end of his career, he had written over 60 books and many hundreds of articles. He died in Geneva, September 16, 1980, one of the most significant psychologists of the twentieth century.

Jean Piaget began his career as a biologist -- specifically, a malacologist! But his interest in science and the history of science soon overtook his interest in snails and clams. As he delved deeper into the thought-processes of doing science, he became interested in the nature of thought itself, especially in the development of thinking. Finding relatively little work done in the area, he had the opportunity to give it a label. He called it **genetic epistemology**, meaning the study of the development of knowledge.

He noticed, for example, that even infants have certain skills in regard to objects in their environment. These skills were certainly simple ones, sensorimotor skills, but they directed the way in which the infant explored his or her environment and so how they gained more knowledge of the world and more sophisticated exploratory skills. These skills he called **schemas**.

For example, an infant knows how to grab his favorite rattle and thrust it into his mouth. He's got that schema down pat. When he comes across some other object -- say daddy's expensive watch, he easily learns to transfer his "grab and thrust" schema to the new object. This Piaget called **assimilation**, specifically assimilating a new object into an old schema.

When our infant comes across another object again -- say a beach ball -- he will try his old schema of grab and thrust. This of course works poorly with the new object. So the schema will adapt to the new object: Perhaps, in this example, "squeeze and drool" would be an appropriate title for the new schema. This is called **accommodation**, specifically accommodating an old schema to a new object.

Assimilation and accommodation are the two sides of **adaptation**, Piaget's term for what most of us would call learning. Piaget saw adaptation, however, as a good deal broader than the kind of learning that Behaviorists in the US were talking about. He saw it as a fundamentally biological process. Even one's grip has to accommodate to a stone, while clay is assimilated into our grip. All living things adapt, even without a nervous system or brain.

Assimilation and accommodation work like pendulum swings at advancing our understanding of the world and our competency in it. According to Piaget, they are directed at a balance between the structure of the mind and the environment, at a certain congruency between the two, that would indicate that you have a good (or at least good-enough) model of the universe. This ideal state he calls **equilibrium**.

As he continued his investigation of children, he noted that there were periods where assimilation dominated, periods where accommodation dominated, and periods of relative equilibrium, and that these periods were similar among all the children he looked at in their nature and their timing. And so he developed the idea of stages of cognitive development. These constitute a lasting contribution to psychology.

[For much more detail on Piaget's theory -- especially the childhood stages -- click here!]

Donald O. Hebb

There are three psychologists who, in my opinion, are most responsible for the development of cognitive psychology as a movement as well as for its incredible popularity today. They are Donald Hebb, George Miller, and Ulric Neisser. There are no doubt others we could add, but I am sure no one would leave these three out!

Donald Olding Hebb was born in 1904 in Chester, Nova Scotia. He graduated from Dalhousie University in 1925, and tried to begin a career as a novelist. He wound up as a school principle in Quebec.

He began as a part-time graduate student at McGill University in Montreal. Here, he began quickly disillusioned with behaviorism and turned to the work of Wolfgang Kohler and Karl Lashley. Working with Lashley on perception in rats, he received his PhD from Harvard in 1936.

He took on a fellowship with Wilder Penfield at the Montreal Neurological Institute, where his research noted that large lesions in the brain often have little effect on a person's perception, thinking, or behavior.

Moving on to Queens University, he researched intelligence testing of animals and humans. He noted that the environment played a far more significant role in intelligence than generally assumed.

In 1942, he worked with Lashley again, this time at the Yerkes Lab of Primate Biology. He then returned to McGill as a professor of psychology, and became the department chairperson in 1948.

The following year, he published his most famous book, **The Organization of Behavior: A Neuropsychological Theory**. This was very well received and made McGill a center for neuropsychology.

The basics of his theory can be summarized by defining three of his terms: First, there is the **Hebb synapse**. Repeated firing of a neuron causes growth or metabolic changes at the synapse that increase the efficiency of that synapse in the future. This is often called **consolidation theory**, and is the most accepted explanation for neural learning today.

Second, there is the **Hebb cell assembly**. There are groups of neurons so interconnected that, once activity begins, it persists well after the original stimulus is gone. Today, people call these **neural nets**.

And third, there is the **phase sequence**. Thinking is what happens when complex sequences of these cell assemblies are activated.

He humbly suggested that his theory is just a new version of **connectionism** -- a neo- or neuro-connectionism. This connectionism is today the basic idea behind most models of neurological functioning. It should be noted that he was president of both the APA and its Canadian cousin, the CPA. Donald Hebb died in 1985.

George A. Miller

George A. Miller, born in 1920, began his career in college as a speech and English major. In 1941, he received his masters in speech from the University of Alabama. In 1946 he received his PhD from Harvard and began to study psycholinguistics.

In 1951, he published his first book, titled **Language and Communication**. In it, he argued that the behaviorist tradition was insufficient to the task of explaining language.

He wrote his most famous paper in 1956: "The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information." In it, he argued that short-term memory could only hold about seven pieces -- called **chunks** -- of information: Seven words, seven numbers, seven faces, whatever. This is still accepted as accurate.

In 1960, Miller founded the Center for Cognitive Studies at Harvard with famous cognitivist developmentalist, Jerome Bruner. In that same year, he published **Plans and the Structure of Behavior** (with Eugene Galanter and Karl Pribram, 1960), which outlined their conception of cognitive psychology. They used the computer as their model of human learning, and used such analogies as information processing, encoding, and retrieval. Miller went so far as to define psychology as the study of the mind, as it had been prior to the behaviorist redefinition of psychology as the study of behavior!

George Miller served as the president of APA 1969, and received the prestigious National Medal of Science in 1991. He is still teaching as professor emeritus at Princeton University.

Ulric Neisser

Ulric Neisser was born in 1928 in Kiel, Germany, and moved with his family to the US at the age of three.

He studied at Harvard as a physics major before switching to psychology. While there, he was influenced by Koffka's work and by George Miller. In 1950, he received his bachelors degree, and in 1956, his PhD. At this point, he was a behaviorist, which was basically what everyone was at the time.

His first teaching position was at Brandeis, where Maslow was department head. Here he was encouraged to pursue his interest in cognition. In 1967, he wrote the book that was to mark the official beginning of the cognitive movement, **Cognitive Psychology**.

Later, in 1976, he wrote **Cognition and Reality**, in which he began to express a dissatisfaction with the linear programming model of cognitive psychology at that time, and the excessive reliance on laboratory work, rather than real-life situations. Over time, he would become a vocal critic of cognitive psychology, and moved towards the environmental psychology of his friend J. J. Gibson.

He is presently at Cornell University where his research interests include memory, especially memory for life events and in natural settings; intelligence, especially individual and group differences in test scores, IQ tests and their social significance; self-concepts, especially as based on self-perception. His latest works include **The Rising Curve: Long-Term Gains in IQ and Related Measures** (1998) and , with L. K. Libby, "Remembering life experiences" (In E. Tulving & F. I .M. Craik's **The Oxford Handbook of Memory**, 2000)

CHAPTER OVERVIEW

6: Women in Psychology

6.1: A Laboratory Study of Fear

A Laboratory Study of Fear: The Case of Peter

Mary Cover Jones (1924)

First published in *Pedagogical Seminary*, *31*, 308-315.

As part of a genetic study of emotions[1], a number of children were observed in order to determine the most effective methods of removing fear responses.

The case of Peter illustrates how a fear may be removed under laboratory conditions. His case was selected from a number of others for the following reasons:

1. Progress in combating the fear reactions was so marked that many of the details of the process could be observed easily.

2. It was possible to continue the study over a period of more than three months.

3. The notes of a running diary show the characteristics of a healthy, normal, interesting child, well adjusted, except for his exaggerated fear reactions. A few descriptive notes show something of his personality:

> *Remarkably active, easily interested, capable of prolonged endeavor . . . . A favorite with the children as well as with the nurses and matrons . . . Peter has a healthy passion for possessions. Everything that he lays his hands on is his. As this is frequently disputed by some other child, there are occasional violent scenes of protest. These disturbances are not more frequent than might be expected in a three-year-old, in view of the fact that he is continually forced to adjust to a large group of children, nor are they more marked in Peter's case than in others of his age. Peter's I.Q. at the age of 2 years and 10 months was 102 on the Kuhlmann Revision of the Binet. At the same time he passed 5 of the 3 year tests on the Stanford Revision. In initiative and constructive ability, however, he is superior to his companions of the same mental age.*

4. The case is a sequel to one recently contributed by Dr. Watson and furnished supplementary material of interest in a genetic study of emotions. Dr. Watson's case illustrated how a fear could be produced experimentally under laboratory conditions[2]. A brief review follows: Albert, eleven months [p. 309] of age, was an infant with a phlegmatic disposition, afraid of nothing "under the sun" except a loud sound made by striking a steel bar. This made him cry. By striking the bar at the same time that Albert touched a white rat, the fear was transferred to the white rat. After seven combined stimulations, rat and sound, Albert not only became greatly disturbed at the sight of a rat, but this fear had spread to include a white rabbit, cotton wool, a fur coat, and the experimenter's hair. It did not transfer to his wooden blocks and other objects very dissimilar to the rat.

In referring to this case, Dr. Watson says, "We have shown experimentally that when you condition a child to show fear of an animal, this fear transfers or spreads in such a way that without separate conditioning he becomes afraid of many animals. If you take any one of these objects producing fear and uncondition, will fear of the other objects in the series disappear at the same time? That is, will the unconditioning spread without further training to other stimuli?"

Dr. Watson intended to continue the study of Albert in an attempt to answer this question, but Albert was removed from the hospital and the series of observations was discontinued.

About three years later this case, which seemed almost to be Albert grown a bit older, was discovered in our laboratory.

Peter was 2 years and 10 months old when we began to study him. He was afraid of a white rat, and this fear extended to a rabbit, a fur coat, a feather, cotton wool, etc., but not to wooden blocks and similar toys. An abridgment of the first laboratory notes on Peter reads as follows:

> *Peter was put in a crib in a play room and immediately became absorbed in his toys. A white rat was introduced into the crib from behind. (The experimenter was behind a screen). At sight of the rat, Peter screamed and fell flat on his back in a paroxysm of fear. The stimulus was removed, and Peter was taken out of the crib and put into a chair. Barbara was brought to the crib and the white rat introduced as before. She exhibited no fear but picked the rat up in her hand. Peter sat quietly watching Barbara and the rat. A string of beads belonging to Peter had been left in the crib. Whenever the rat touched a part of the string he would say "my beads" in a complaining voice, although he made no objections when Barbara touched them. Invited to get down from the chair, he shook his head, fear not yet subsided. Twenty-five minutes elapsed before he was ready to play about freely.*

The next day his reactions to the following situations and objects were noted: [p. 310]

Play room and crib . . . . Selected toys, got into crib without protest
White ball rolled in . . . . Picked it up and held it
Fur rug hung over crib . . . . .Cried until it was removed
Fur coat hung over crib .Cried until it was removed
Cotton . . . . Whimpered, withdrew, cried
Hat with feathers . . . . Cried
Blue woolly sweater . . . . Looked, turned away, no fear
White toy rabbit of rough cloth . . . . No interest, no fear
Wooden doll . . . . No interest, no fear

This case made it possible for the experiment to continue where Dr. Watson had left off. The first problem was that of "unconditioning" a fear response to an animal, and the second, that of determining whether unconditioning to one stimulus spreads without further training to other stimuli.

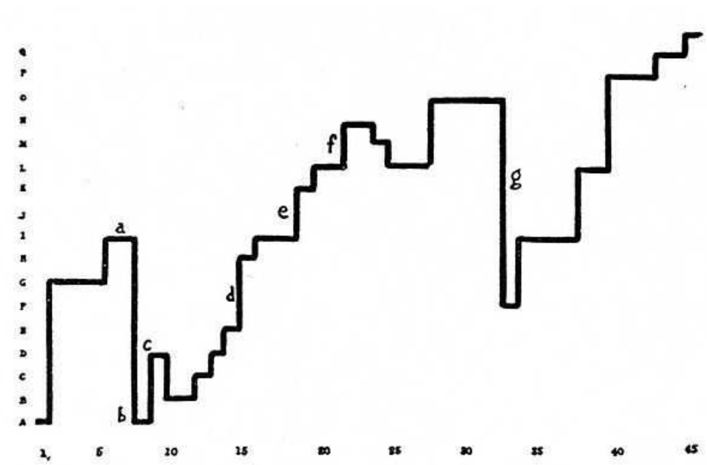
From the test situations which were used to reveal fears, it was found that Peter showed even more marked fear responses to the rabbit than to the rat. It was decided to use the rabbit for unconditioning and to proceed as follows: Each day Peter and three other children were brought to the laboratory for a play period. The other children were selected carefully because of their entirely fearless attitude toward the rabbit and because of their satisfactory adjustments in general. The rabbit was always present during a part of the play period. From to time Peter was brought in alone so that his reactions could be observed and progress noted.

From reading over the notes for each session it was apparent that there had been improvement by more or less regular steps from almost complete terror at sight of the rabbit to a completely positive response with no signs of disturbance. New situations requiring closer contact with the rabbit had been gradually introduced and the degree to which these situations were avoided, tolerated, or welcomed, at each experimental session, gave the measure of improvement. Analysis of the notes on Peter's reactions indicated the following progressive steps in his degrees of toleration:

A. Rabbit anywhere in the room in a cage causes fear reactions.
B. " 12 feet away in cage tolerated.
C. " 4 " " " " "
D. " 3 " " " " "
E. " close " " "
F. " free in room tolerated.
G. " touched when experimenter holds it.
H. " touched when free in room.
I. " defied by spitting at it, throwing things at it, imitating it. [p. 311]
J. Rabbit allowed on tray of high chair.
K. Squats in defenseless position beside rabbit.

L. Helps experimenter to carry rabbit to its cage.
M. Holds rabbit on lap.
N. Stays alone in room with rabbit.
O. Allows rabbit in play pen with him.
P. Fondles rabbit affectionately.
Q. Lets rabbit nibble his fingers.

These "degrees of toleration" merely represented the stages in which improvement occurred. They did not give any indications of the intervals between steps, nor of the plateaus, relapses, and sudden gains which were actually evident. To show these features a curve was drawn by using the seventeen steps given above as the Y axis of a chart and the experimental sessions as the X axis. The units are not equal on either axis, as the "degrees of toleration" have merely been set down as they appeared from consideration of the laboratory notes with no attempt to evaluate the steps. Likewise the experimental sessions were not equi-distant in time. Peter was seen twice daily for a period and thence only once a day. At one point illness and quarantine interrupted the experiments for two months. There is no indication of these irregularities on the chart. For example, along the X axis, 1 represents the date December 4th when the observation began. 11 and 12 represent the dates March 10 A.M. and P.M. (from December 17 to March 7, Peter was not available for study).



[p. 312]

The question arose as to whether or not the points on the Y axis which indicated progress to the experimenter represented real advance and not merely idiosyncratic reactions of the subject. The "tolerance series" as indicated by the experimenter was presented in random order to six graduate students and instructors in psychology to be arranged so as to indicate increase in tolerance, in their judgment. An average correlation of .70 with the experimenter's arrangement was found for the six ratings. This indicates that the experimenter was justified from an a priori point of view in designating the steps to be progressive stages.

The first seven periods show how Peter progressed from a great fear of the rabbit to a tranquil indifference and even a voluntary pat on the rabbit's back when others were setting the example. The notes for the seventh period (see a on chart) read:

> *Laurel, Mary, Arthur, Peter playing together in the laboratory. Experimenter put rabbit down on floor. Arthur said, "Peter doesn't cry when he sees the rabbit come out." Peter, "No." He was a little concerned as to whether or not the rabbit would eat his kiddie car. Laurel and Mary stroked the rabbit and chattered away excitedly. Peter walked over, touched the rabbit on the back, exulting, "I touched him on the end."*

At this period Peter was taken to the hospital with scarlet fever. He did not return for two months.

By referring to the chart at (b), it will be noted that the line shows a decided drop to the early level of fear reaction when he returned. This was easily explained by the nurse who brought Peter from the hospital. As they were entering a taxi at the door of the hospital, a large dog, running past, jumped at them. Both Peter and the nurse were very much frightened, Peter so much that he lay in the taxi pale and quiet, and the nurse debated whether or not to return him to the hospital. This seemed reason enough for his precipitate descent back to the original fear level. Being threatened by a large dog when ill, and in a strange place and being with an adult who also showed fear, was a terrifying situation against which our training could not have fortified him.

At this point (b) we began another method of treatment, that of "direct conditioning." Peter was seated in a high chair and given food which he liked. The experimenter brought the rabbit in a wire cage as close as she could without arousing a response which would interfere with the eating. [p. 313]

Through the presence of the pleasant stimulus (food) whenever the rabbit was shown, the fear was eliminated gradually in favor of a positive response. Occasionally also, other children were brought in to help with the "unconditioning." These facts are of interest in following the charted progress. The first decided rise at (c) was due to the presence of another child who influenced Peter's reaction. The notes for this day read:

> *Lawrence and Peter sitting near together in their high chairs eating candy. Rabbit in cage put down 12 feet away. Peter began to cry. Lawrence said, "Oh, rabbit." Clambered down, ran over and looked in the cage at him. Peter followed close and watched.*

The next two decided rises at (d) and (e) occurred on the day when a student assistant, Dr. S., was present. Peter was very fond of Dr. S. whom he insisted was his "papa." Although Dr. S. did not directly influence Peter by any overt suggestions, it may be that having him there contributed to Peter's general feeling of well being and thus indirectly affected his reactions. The fourth rise on the chart at (f) was, like the first, due to the influence of another child. Notes for the 21st session read:

> *Peter with candy in high chair. Experimenter brought rabbit and sat down in front of the tray with it. Peter cried out, "I don't want him," and withdrew. Rabbit was given to another child sitting near to hold. His holding the rabbit served as a powerful suggestion; Peter wanted the rabbit on his lap, and held it for an instant.*

The decided drop at (g) was caused by a slight scratch when Peter was helping to carry the rabbit to his cage. The rapid ascent following shows how quickly he regained lost ground.

In one of our last sessions, Peter showed no fear although another child was present who showed marked disturbance at sight of the rabbit.

An attempt was made from time to time to see what verbal organization accompanied this process of "unconditioning." Upon Peter's return from the hospital, the following conversation took place:

E.: (experimenter) What do you do upstairs, Peter? (The laboratory was upstairs).
P.: I see my brother. Take me up to see my brother.
E.: What else will you see?
P.: Blocks. [p. 314]

Peter's reference to blocks indicated a definite memory as he played with blocks only in the laboratory. No further response of any significance could be elicited. In the laboratory two days later (he had seen the rabbit once in the meantime), he said suddenly, "Beads can't bite me, beads can only look at me." Toward the end of the training an occasional "I like the rabbit," was all the language he had to parallel the changed emotional organization.

Early in the experiment an attempt was made to get some measure of the visceral changes accompanying Peter's fear reactions. On one occasion Dr. S. determined Peter's blood pressure outside the laboratory and again later, in the laboratory while he was in a state of much anxiety caused by the rabbit's being held close to him by the experimenter. The diastolic blood pressure changed from 65 to 80 on this occasion. Peter was taken to the infirmary the next day for the routine physical examination and developed there a suspicion of medical instruments which made it inadvisable to proceed with this phase of the work.

Peter has gone home to a difficult environment but the experimenter is still in touch with him. He showed in the last interview, as on the later portions of the chart, a genuine fondness for the rabbit. What has happened to the fear of the other objects? The fear of the cotton, the fur coat, feathers, was entirely absent at our last interview. He looked at them, handled them, and immediately turned to something which interested him more. The reaction to the rats, and the fur rug with the stuffed head was greatly modified and improved. While he did not show the fondness for these that was apparent with the rabbit, he had made a fair adjustment. For example, Peter would pick up the tin box containing frogs or rats and carry it around the room. When requested, he picked up the fur rug and carried it to the experimenter.

What would Peter do if confronted by a strange animal? At the last interview the experimenter presented a mouse and a tangled mass of angleworms. At first sight, Peter showed slight distress reactions and moved away, but before the period was over he was carrying the worms about and watching the mouse with undisturbed interest. By "unconditioning" Peter to the rabbit, he has apparently been helped to overcome many superfluous fears, some completely, some to a less degree. His tolerance of strange animals and unfamiliar situations has apparently increased. [p. 315]

The study is still incomplete. Peter's fear of the animals which were shown him was probably not a directly conditioned fear. It is unlikely that he had ever had any experience with white rats, for example. Where the fear originated and with what stimulus, is not known. Nor is it known what Peter would do if he were again confronted with the original fear situation. All of the fears which were "unconditioned" were transferred fears, and it has not yet been learned whether or not the primary fear can be eliminated by training the transfers.

Another matter which must be left to speculation is the future welfare of the subject. His "home" consists of one furnished room which is occupied by his mother and father, a brother of nine years and himself. Since the death of an older sister, he is the recipient of most of the unwise affection of his parents. His brother appears to bear him a grudge because of this favoritism, as might be expected. Peter hears continually, "Ben is so bad and so dumb, but Peter is so good and so smart!" His mother is a highly emotional individual who can not get through an interview, however brief, without a display of tears. She is totally incapable of providing a home on the $25 a week which her husband steadily earns. In an attempt to control Peter she resorts to frequent fear suggestions. "Come in Peter, some one wants to steal you." To her erratic resorts to discipline, Peter reacts with temper tantrums. He was denied a summer in the country because his father "forgets he's tired when he has Peter around." Surely a discouraging outlook for Peter.

But the recent development of psychological studies of young children and the growing tendency to carry the knowledge gained in the psychological laboratories into the home and school induce us to predict a more wholesome treatment of a future generation of Peters.

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Accepted for publication by
JOHN B. WATSON

Footnotes

[1] The research was conducted with the advice of Dr. John B. Watson, by means of a subvention granted by the Laura Spelman Rockefeller Memorial to the Institute of Educational Research of Teachers' College.

[2] Watson, J. B., and R. R. Studies in Infant Psychology, Scientific Monthly, December 1921.

6.2: Placing Women in the History of Psychology

Read this!

Placing Women in the History of Psychology

6.3: Women of Color in Psychology

Explore this!

Women of Color in Psychology

CHAPTER OVERVIEW

7: Mental Illness/Clinical Psychology

7.1: Clinical Psychology

CLINICAL PSYCHOLOGY
Lightner Witmer (1907)

First published in *Psychological Clinic, 1*, 1-9

During the last ten years the laboratory of psychology at the University of Pennsylvania has conducted, under my direction, what I have called "a psychological clinic". Children from the public schools of Philadelphia and adjacent cities have been brought to the laboratory by parents or teachers; these children had made themselves conspicuous because of an inability to progress in school work as rapidly as other children, or because of moral defects which rendered them difficult to manage under ordinary discipline.

When brought to the psychological clinic, such children are given a physical and mental examination; if the result of this examination shows it to be desirable, they are then sent to specialists for the eye or ear, for the nose and throat, and for nervous diseases, one or all, as each case may require. The result of this conjoint medical and psychological examination is a diagnosis of the child's mental and physical condition and the recommendation of appropriate medical and pedagogical treatment. The progress of some of these children has been followed for a term of years.

To illustrate the operation of the psychological clinic, take a recent case sent to the laboratory from a city of Pennsylvania, not far from Philadelphia. The child was brought by his parents, on the recommendation of the Superintendent of Schools. Examination revealed a boy ten years of age, without apparent physical defect, who had spent four years at school, but had made so little progress that his ignorance of the printed symbols of the alphabet made it necessary to use the illiterate card to test his vision. Nothing in the child's heredity or early history revealed any ground for the suspicion of degeneracy, nor did the child's physical appearance warrant this diagnosis. The boy appeared to be of normal intelligence, except for the retardation in school work. The examination of the neurologist to whom he was sent, Dr. William G. Spiller, confirmed the absence of conspicuous mental degeneracy and of physical defect. The oculist, Dr. William C. Posey, found nothing more serious than a slight far-sighted astigmatism, and the examination of Dr. George C. Stout for adenoids, gave the child a clean bill of health, so far as the nose and pharynx were concerned. On the conclusion of this examination he was, necessarily, returned to the school from which he came, with the recommendation to the teacher of a course of treatment to develop the child's intelligence. It will require at least three months' observation to determine whether his present pedagogical retardation is based upon an arrest of cerebral development or is merely the result of inadequate methods of education. This case is unequivocally one for the psychologist.

My attention was first drawn to the phenomena of retardation in the year 1889. At that time, while a student of psychology at the University of Pennsylvania, I had charge of the English branches in a college preparatory school of Philadelphia. In my classes at this academy I was called upon to give instruction in English to a boy preparing for entrance to college, who showed a remarkable deficiency in the English language. His compositions seldom contained a single sentence that had been correctly formed. For example, there was little or no distinction between the present and the past tenses of verbs; the endings of many words were clipped off, and this was especially noticeable in those words in which a final ending distinguished the plural from the singular, or an adverb from an adjective. As it seemed doubtful whether he would ever be able to enter college without special instruction in English, I was engaged to tutor him in the English branches.

I had no sooner undertaken this work than I saw the necessity of beginning with the elements of language and teaching him as one would teach a boy, say, in the third grade. Before long I discovered that I must start still further back. I had found it impossible, through oral and written exercises, to fix in his mind the elementary forms of words as parts of speech in a sentence. This seemed to be owing to the fact that he had verbal deafness. He was quite able to hear even a faint sound, like the ticking of a watch, but he could not hear the difference in the sound of such words as *grasp* and *grasped* . This verbal deafness was associated with, and I now believe was probably caused by, a defect of articulation. Thus the boy's written language was a fairly exact replica of his spoken language; and he probably heard the words that others spoke as he himself spoke them. I therefore undertook to give him an elementary training in articulation to remedy the defects which are ordinarily corrected, through imitation, by the time a child is three or four years old. I gave practically no attention to the subjects required in English for college entrance, spending all my time on the drill in articulation and in perfecting his verbal audition and teaching him the simplest elements of written language. The result was a great improvement in all his written work, and he succeeded in entering the college department of the University of Pennsylvania in the following year.

In 1894-95, I found him as a college student in my classes at the University of Pennsylvania. His articulation, his written discourse and his verbal audition were very deficient for a boy of his years. In consequence he was unable to acquire the technical terminology of my branch, and I have no doubt that he passed very few examinations excepting through the sympathy of his instructors who overlooked the serious imperfections of his written work, owing to the fact that he was in other respects a fair student. When it came to the final examinations for the bachelor's degree, however, he failed and was compelled to repeat much of the work of his senior year. He subsequently entered and graduated from one of the professional departments of the University. His deficiencies in language, I believe, have never been entirely overcome.

I felt very keenly how much this boy was losing through his speech defect. His school work, his college course, and doubtless his professional career were all seriously hampered. I was confident at the time, and this confidence has been justified by subsequent experience with similar cases, that if he had been given adequate instruction in articulation in the early years of childhood, he could have overcome his defect. With the improvement in articulation there would have come an improved power of apprehending spoken and written language. That nothing was done for him in the early years, nor indeed at any time, excepting for the brief period of private instruction in English and some lessons in elocution, is remarkable, for the speech defect was primarily owing to an injury to the head in the second year of life, and his father was a physician who might have been expected to appreciate the necessity of special training in a case of retardation caused by a brain injury.

The second case to attract my interest was a boy fourteen years of age, who was brought to the laboratory of psychology by his grade teacher. He was one of those children of great interest to the teacher, known to the profession as a chronic bad speller. His teacher, Miss Margaret T. Maguire, now the supervising principal of a grammar school of Philadelphia, was at that time a student of psychology at the University of Pennsylvania; she was imbued with the idea that a psychologist should be able, through examination, to ascertain the causes of a deficiency in spelling and to recommend the appropriate pedagogical treatment for its amelioration or cure.

With this case, in March, 1896, the work of the psychological clinic was begun. At that time I could not find that the science of psychology had ever addressed itself to the ascertainment of the causes and treatment of a deficiency in spelling. Yet here was a simple developmental defect of memory; and memory is a mental process of which the science of psychology is supposed to furnish the only authoritative knowledge. It appeared to me that if psychology was worth anything to me or to others it should be able to assist the efforts of a teacher in a retarded case of this kind.

"The final test of the value of what is called science is its applicability" are words quoted from the recent address of the President of the American Association for the Advancement of Science. With Huxley and President Woodward, I believe that there is no valid distinction between a pure science and an applied science. The practical needs of the astronomer to eliminate the personal equation from his observations led to the invention of the chronograph and the chronoscope. Without these two instruments, modern psychology and physiology could not possibly have achieved the results of the last fifty years. If Helmholtz had not made the chronograph an instrument of precision in physiology and psychology; if Fechner had not lifted a weight to determine the threshold of sensory discrimination, the field of scientific work represented to-day by clinical psychology could never have been developed. The pure and the applied sciences advance in a single front. What retards the progress of one, retards the progress of the other; what fosters one, fosters the other. But in the final analysis the progress of psychology, as of every other science, will be determined by the value and amount of its contributions to the advancement of the human race.

The absence of any principles to guide me made it necessary to apply myself directly to the study of these children, working out my methods as I went along. In the spring of 1896 I saw several other cases of children suffering from the retardation of some special function, like that of spelling, or from general retardation, and I undertook the training of these children for a certain number of hours each week. Since that time the psychological clinic has been regularly conducted in connection with the laboratory of psychology at the University of Pennsylvania. The study of these cases has also formed a regular part of the instruction offered to students in child psychology.

In December, 1896, I outlined in an address delivered before the American Psychological Association a scheme of practical work in psychology. The proposed plan of organization comprised:

- The investigation of the phenomena of mental development in school children, as manifested more particularly in mental and moral retardation, by means of the statistical and clinical methods.
- A psychological clinic, supplemented by a training school in the nature of a hospital school, for the treatment of all classes of children suffering from retardation or physical defects interfering with school progress.
- The offering of practical work to those engaged in the professions of teaching and medicine, and to those interested in social work, in the observation and training of normal and retarded children.
- The training of students for a new profession-that of the psychological expert, who should find his career in connection with the school system, through the examination and treatment of mentally and morally retarded children, or in connection with the

practice of medicine.

In the summer of 1897 the department of psychology in the University of Pennsylvania was able to put the larger part of this plan into operation. A four weeks' course was given under the auspices of the American Society for the Extension of University Teaching. In addition to lecture and laboratory courses in experimental and physiological psychology, a course in child psychology was given to demonstrate the various methods of child psychology, but especially the clinical method. The psychological clinic was conducted daily, and a training school was in operation in which a number of children were under the daily instruction of Miss Mary E. Marvin. At the clinic, cases were presented of children suffering from defects of the eye, the ear, deficiency in motor ability, or in memory and attention; and in the training school, children were taught throughout the session of the Summer School, receiving pedagogical treatment for the cure of stammering and other speech defects, for defects of written language (such as bad spelling), and for motor defects.

From that time until the present I have continued the examination and treatment of children in the psychological clinic. The number of cases seen each week has been limited, because the means were not at hand for satisfactorily treating a large number of cases. I felt, also, that before offering to treat these children on a large scale I needed some years of experience and extensive study, which could only be obtained through the prolonged observation of a few cases. Above all, I appreciated the great necessity of training a group of students upon whose assistance I could rely. The time has now come for a wider development of this work. To further this object and to provide for the adequate publication of the results that are being obtained in this new field of psychological investigation, it was determined to found this journal, *The Psychological Clinic*.

My own preparation for the work has been facilitated through my connection as consulting psychologist with the Pennsylvania Training School for Feeble-Minded Children at Elwyn, and a similar connection with the Haddonfield Training School and Miss Marvin's Home School in West Philadelphia.

Clinical psychology is naturally very closely related to medicine. At the very beginning of my work I was much encouraged by the appreciation of the late Provost of the University of Pennsylvania, Dr. William Pepper, who at one time proposed to establish a psychological laboratory in connection with the William Pepper Clinical Laboratory of Medicine. At his suggestion, psychology was made an elective branch in what was then the newly organized fourth year of the course in medicine. At a subsequent reorganization of the medical course, however, it was found necessary to drop the subject from the curriculum.

I also desire to acknowledge my obligation to Dr. S. Weir Mitchell for co-operation in the examination of a number of cases and for his constant interest in this line of investigation. I have also enjoyed the similar co-operation of Dr. Charles K. Mills, Dr. William G. Spiller, the late Dr. Harrison Allen, Dr. Alfred Stengel, Dr. William Campbell Posey, Dr. George C. Stout, and Dr. Joseph Collins, of New York. Dr. Collins will continue this co-operation as an associate editor of *The Psychological Clinic*.

The appreciation of the relation of psychology to the practice of medicine in general, and to psychiatry in particular, has been of slow growth. The first intelligent treatment of the insane was accorded by Pinel in the latter part of the eighteenth century, a century that was marked by the rapid development of the science of psychology, and which brought forth the work of Pereire in teaching oral speech to the deaf, and the "Emile" of Rousseau. A few medical men have had a natural aptitude for psychological analysis. From them has come the chief development of the medical aspects of psychology,-from Seguin and Charcot in France, Carpenter and Maudsley in England, and Weir Mitchell in this country. Psychological insight will carry the physician or teacher far on the road to professional achievement, but at the present day the necessity for a more definite acquaintance with psychological method and facts is strongly felt. It is noteworthy that perhaps the most prominent name connected with psychiatry to-day is that of Kraepelin, who was among the first to seek the training in experimental psychology afforded by the newly established laboratory at Leipzig.

Although clinical psychology is closely related to medicine, it is quite as closely related to sociology and to pedagogy. The school room, the juvenile court, and the streets are a larger laboratory of psychology. An abundance of material for scientific study fails to be utilized, because the interest of psychologists is elsewhere engaged, and those in constant touch with the actual phenomena do not possess the training necessary to make their experience and observation of scientific value.

While the field of clinical psychology is to some extent occupied by the physician, especially by the psychiatrist, and while I expect to rely in a great measure upon the educator and social worker for the more important contributions to this branch of psychology, it is nevertheless true that none of these has quite the training necessary for this kind of work. For that matter, neither has the psychologist, unless he has acquired this training from other sources than the usual course of instruction in psychology. In fact, we must look forward to the training of men to a new profession which will be exercised more particularly in connection with educational problems, but for which the training of the psychologist will be a prerequisite.

For this reason not a small part of the work of the laboratory of psychology in the University of Pennsylvania for the past ten years has been devoted to the training of students in child psychology, and especially in the clinical method. The greater number of these students have been actively engaged in the profession of teaching. Important contributions to psychology and pedagogy, the publication of which in the form of monographs has already been begun, will serve to demonstrate that original research of value can be carried on by those who are actively engaged in educational or other professional work. There have been associated in this work of the laboratory of psychology, Superintendent Twitmyer, of Wilmington; Superintendent Bryan, of Camden; District Superintendent Cornman, of Philadelphia; Mr. J. M. McCallie, Supervising Principal of the Trenton Schools; Mr. Edward A. Huntington, Principal of a Special School in Philadelphia; Miss Clara H. Town, Resident Psychologist at the Friends' Asylum for the Insane, and a number of special teachers for the blind, the deaf, and mentally deficient children. I did not venture to begin the publication of this journal until I felt assured of the assistance of a number of fellow-workers in clinical psychology as contributors to the journal. As this work has grown up in the neighborhood of Philadelphia, it is probable that a greater number of students, equipped to carry on the work of clinical psychology, may be found in this neighborhood than elsewhere, but it is hoped that this journal will have a wider influence, and that the co-operation of those who are developing clinical psychology throughout the country will be extended the journal.

The phraseology of "clinical psychology" and "psychological clinic" will doubtless strike many as an odd juxtaposition of terms relating to quite disparate subjects. While the term "clinical" has been borrowed from medicine, clinical psychology is not a medical psychology. I have borrowed the word "clinical" from medicine, because it is the best term I can find to indicate the character of the method which I deem necessary for this work. Words seldom retain their original significance, and clinical medicine, is not what the word implies,-the work of a practicing physician at the bedside of a patient. The term "clinical" implies a method, and not a locality. When the clinical method in medicine was established on a scientific basis, mainly through the efforts of Boerhaave at the University of Leiden, its development came in response to a revolt against the philosophical and didactic methods that more or less dominated medicine up to that time. Clinical psychology likewise is a protestant against a psychology that derives psychological and pedagogical principles from philosophical speculations and against a psychology that applies the results of laboratory experimentation directly to children in the school room.

The teacher's interest is and should be directed to the subjects which comprise the curriculum, and which he wishes to impress upon the minds of the children assigned to his care. It is not what *the child is*, but *what he should be taught* that occupies the center of his attention. Pedagogy is primarily devoted to mass instruction, that is, teaching the subjects of the curriculum to classes of children without reference to the individual differences presented by the members of a class. The clinical psychologist is interested primarily in the individual child. As the physician examines his patient and proposes treatment with a definite purpose in view, namely the patient's cure, so the clinical psychologist examines a child with a single definite object in view,-the next step in the child's mental and physical development. It is here that the relation between science and practice becomes worthy of discrimination. The physician *may* have solely in mind the cure of his patient, but if he is to be more than a mere practitioner and to contribute to the advance of medicine, he will look upon his efforts as an experiment, every feature of which must indeed have a definite purpose,-the cure of the patient,-but he will study every favorable or unfavorable reaction of the patient with reference to the patient's previous condition and the remedial agents he has employed. In the same way the purpose of the clinical psychologist, as a contributor to science, is to discover the relation between cause and effect in applying the various pedagogical remedies to a child who is suffering from general or special retardation.

I would not have it thought that the method of clinical psychology is limited necessarily to mentally and morally retarded children. These children are not, properly speaking, abnormal, nor is the condition of many of them to be designated as in any way pathological. They deviate from the average of children only in being at a lower stage of individual development. Clinical psychology, therefore, does not exclude from consideration other types of children that deviate from the average-for example, the precocious child and the genius. Indeed, the clinical method is applicable even to the so-called normal child. For the methods of clinical psychology are necessarily invoked wherever the status of an individual mind is determined by observation and experiment, and pedagogical treatment applied to effect a change, i.e., the development of such individual mind. Whether the subject be a child or an adult, the examination and treatment may be conducted and their results expressed in the terms of the clinical method.

7.2: On Being Sane in Insane Places

https://www.weber.edu/wsuimages/psychology/FacultySites/Horvat/OnBeingSaneInInsanePlaces.PDF

8: Developmental/Educational Psychology

8.1: Mamie P. Clark’s denied research

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Mamie P. Clark's Denied Research

8.2: The History of School Psychology- Understanding the Past to Not Repeat It

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Read starting on page 37 (Major Movements Related to the Field of School Psychology) through page 59 (Sample Research Activities Reported from a University-School Training Partnership)

The History of School Psychology: Understanding the Past to Not Repeat It

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